The Riemann Hypothesis by Exclusion of the Prime-Trace Amplitude Separator

A Full Audit-Grade AASC Endpoint-Rigidity Proof via Explicit-Formula Scale Discrimination

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Abstract

This manuscript gives a full audit-grade AASC endpoint-exclusion proof of the Riemann Hypothesis through the prime-trace amplitude route. It does not introduce a new zero-free region, zero-density estimate, positivity criterion, Hilbert–Polya operator, random-matrix theorem, or numerical verification. It uses the standard explicit-formula zero-to-prime-trace correspondence as the official endpoint route on the fixed zeta/prime-trace carrier.

Let

$$A_\zeta(\rho) :\Longleftrightarrow \zeta(\rho) = 0 \ \wedge \ 0 < \mathrm{Re}(\rho) < 1$$

be nontrivial zeta zerohood, and let

$$L_{\mathrm{crit}}(\rho) :\Longleftrightarrow \mathrm{Re}(\rho) = \frac{1}{2}$$

be critical-line readout. The positive endpoint is

$$\mathrm{RH}^+ := \forall \rho \, (A_\zeta(\rho) \Rightarrow L_{\mathrm{crit}}(\rho)).$$

The bare negative branch is the ordinary off-line zerohood assertion

$$\mathrm{RH}^-_{\mathrm{bare}} := \exists \rho \, (A_\zeta(\rho) \wedge \neg L_{\mathrm{crit}}(\rho)).$$

Descriptive off-line coordinate reports are preserved. The contradiction below is not derived from $\mathrm{Re}(\rho) \neq 1/2$ alone.

Front-loaded dependency lock. This manuscript has one nonclassical dependency: official endpoint-defeating counterexample force is theorem-bearing endpoint use. Once that is granted, the remaining proof is target instantiation plus route exhaustion. The dependency is not hidden in the explicit formula, in the zeta carrier, in the critical-normalized amplitude interface, or in the local reductio rule. Bare negative content may exist as syntax, hypothesis, support, or ordinary theoremhood; but when it is used to defeat the official endpoint on the same fixed carrier, that use fixes target, carrier, branch, reportability, downstream reuse, positive-branch exclusion, act-time finality, and lawful-redescription stability. That is endpoint use. A reader who denies this dependency should attack the no-autonomous-counterexample theorem directly; the rest of the manuscript is the RH-EF instantiation of that dependency.

The decisive transport is through the explicit formula and the structural-to-explicit-formula equivalence bridge. The structural RH role decomposition supplies analytic zerohood, carrier preservation, kernel-governed report identity, carrier-preservation bridge, and critical-line readout. The present manuscript instantiates that bridge as B_ζ^{EF}, the explicit-formula amplitude bridge, which does not contain line support by definition. It binds same-zero zerohood to an explicit-formula contribution, line readout to neutral critical-normalized order, and off-line readout to non-neutral order. For a zero $\rho = \beta + i\gamma$, the zero contribution carries scale x^ρ/ρ in a standard Chebyshev prime trace. Under critical normalization by $x^{1/2}$, its amplitude profile is governed by $x^{\beta-1/2}$. Critical-line zeros occupy the neutral amplitude interface x^0. An off-line zero supplies a non-neutral scale displacement. When this displacement is used as same-carrier endpoint counterforce against the official RH endpoint through the explicit-formula route, it is not merely a native coordinate report; it is endpoint-governing prime-trace amplitude separator occupation. Native-counterforce exhaustion then proves that ordinary native off-line amplitude falsification is not a fifth endpoint role: endpoint-standing non-neutral order is bridge-neutralized, support-only, domain/interface shift,

or independent amplitude discriminator work. In the live off-line countercase, bridge-neutralization is unavailable, support-only use lacks endpoint counterforce, and fixed-domain evaluation excludes domain/interface shift; the remaining branch is $D_{\mathrm{amp}}(\rho)$.

The proof uses the AASC kernel, UEAP report discipline, and ATS anchor–tensor–skin decomposition as structural locking support. UEAP prevents report laundering from coordinate or support status into endpoint status. ATS fixes the zeta/prime-trace anchors, identifies the amplitude exponent as tensor content rather than notation skin, and blocks surrogate or normalization skins from carrying endpoint force. The AMetric and no-independent-discriminator consequences of endpoint use exclude independent same-domain amplitude endpoint discriminators. The live exact-complement local countercase $[\neg L_{\mathrm{crit}}(\rho)]_i$ induces local amplitude separator occupation; that occupation induces $D_{\mathrm{amp}}(\rho)$, while the same endpoint use yields $N_{\mathrm{amp}}(\rho) \Rightarrow \neg D_{\mathrm{amp}}(\rho)$. The local countercase discharges and gives $L_{\mathrm{crit}}(\rho)$ for arbitrary nontrivial zerohood. Universal generalization gives the Riemann Hypothesis.

Mathematical-status lock. AASC is used here as a mathematical constraint formalism. The proof is a theorem chain about endpoint use, target instantiation, explicit-formula amplitude projection, role occupation, and no-independent-discriminator closure. It is not philosophical commentary or an optional interpretive overlay.

Proof-class lock. The proof class is *kernel-forced prime-trace amplitude separator exclusion*. Wrong-mode demands for a zero-free region, density estimate, spectral operator, positivity criterion, or numerical extension do not touch the proof class unless they identify a failure in target instantiation, model adequacy, amplitude projection, separator induction, or no-independent closure.

Corpus-support boundary. The AASC Corpus Control Matrix is used only to select structural support anchors for kernel forcing, UEAP report discipline, ATS role decomposition, AMetric no-selector discipline, role-occupancy closure, no post-selection repair, silent-redescription exclusion, and same-domain incompleteness control. Those corpus anchors do not supply analytic zero-location facts. They are summarized in Appendix E.

Lean support boundary. This manuscript remains the mathematical proof. The Lean4 material in Appendix A is audit support for the RH endpoint-role surface: explicit-formula route objects, counterexample-force use, amplitude separator occupation, discriminator induction, no-independent closure, and pointwise critical-line closeout. It is not a substitute for the manuscript proof and does not claim a first-principles Lean formalization of analytic number theory.

Front-Loaded Dependency Lock

This manuscript has one nonclassical dependency: official endpoint-defeating counterexample force is theorem-bearing endpoint use. Once that is granted, the remaining proof is target instantiation plus route exhaustion. This sentence is the dependency lock for the whole paper. The proof does not ask the reader to accept a new analytic estimate, a hidden positivity principle, a disguised zero-free region, or a redefinition of the zeta carrier. It asks the reader to accept the endpoint-use thesis for official counterexample force, then verifies the RH-EF target instantiation and exhausts the possible roles of the projected off-line amplitude branch.

The lock has two sides. First, raw counterexample content is not endpoint use: an object ρ, the formula $A_\zeta(\rho) \wedge \neg L_{\mathrm{crit}}(\rho)$, and the non-neutral order statement $\mathrm{ord}_{1/2}(\rho) \neq 0$ may remain syntax, hypothesis, descriptive analytic content, support, or ordinary theoremhood. Second, endpoint-defeating counterexample force is not merely an object: it is the mathematical act of offering that object or theorem as the same-carrier defeat of the official endpoint. Such an act fixes the target, carrier, branch, exclusion of the positive endpoint, reportability, downstream reuse, act-time status, and lawful-redescription role. Those are exactly the endpoint-use features governed by the kernel and the A+ consequence layer.

After this dependency is stated, the rest of the proof has no hidden nonclassical step. The RH target is instantiated through the explicit-formula route; the structural RH roles are bridged to critical-normalized prime-trace order; native off-line counterforce is exhausted into bridge-neutralized, support-only, bookkeeping, domain/interface-shift, or independent-amplitude-discriminator roles; the live fixed-domain subproof eliminates the first four; and A+ excludes the remaining discriminator. Therefore a same-mode objection should not scatter across the manuscript. It should either deny the endpoint-use status of official counterexample force, or identify a failure in target instantiation, role projection, native-counterforce exhaustion, local discharge, or A+ closure.

Claim-status ledger

The following ledger fixes report levels before the proof begins. It is included to prevent carrier, support, endpoint use, and audit support from being read as interchangeable roles.

Layer	**RH-EF statement**	**Status**
Carrier layer	Zeta carrier, nontrivial zerohood, prime-trace carrier, explicit-formula route, same-zero identity, and critical normalization.	Context and adequacy; not RH and not line support.
Standing layer	$A_\zeta(\rho)$.	Zerohood standing for a fixed point ρ.
Positive endpoint image	$L_{\mathrm{crit}}(\rho)$, equivalently $\mathrm{ord}_{1/2}(\rho) = 0$ and $x^{\beta-1/2} = x^0$.	Endpoint readout obtained only after the local countercase is excluded.
Raw negative content	$A_\zeta(\rho) \wedge \neg L_{\mathrm{crit}}(\rho)$.	Lawful raw, support, local-hypothesis, or ordinary theorem content.
Explicit-formula support	x^ρ/ρ, $x^{\beta-1/2}$, and non-neutral order.	Lawful analytic support unless deployed with endpoint-defeating force.
Native off-line falsification package	$O_{\mathrm{nat}}(\rho)$ deployed against the official RH endpoint.	Governed endpoint use; route-exhausted into bridge-neutralized, support/bookkeeping, domain shift, or $D_{\mathrm{amp}}(\rho)$.
Surrogate analytic programs	Zero-free regions, density estimates, positivity criteria, spectral models, smoothing regimes, and numerical verification.	Support or bridge programs until they supply the fixed RH-EF endpoint bridge.
Lean layer	RH endpoint-role audit surface.	Reproducibility support for the proof spine, not first-principles analytic number theory.

Contents

Read This First: Endpoint Proof Spine

Let $\rho \in \mathbb{C}$ be arbitrary, and write $\rho = \beta + i\gamma$. The proof works locally under the zerohood assumption $A_\zeta(\rho)$. The raw local split is

$$A_\zeta(\rho) \Rightarrow \big(L_{\mathrm{crit}}(\rho) \vee \neg L_{\mathrm{crit}}(\rho)\big).$$

Open the exact-complement local countercase

$$[\neg L_{\mathrm{crit}}(\rho)]_i.$$

Under the official explicit-formula endpoint route, $A_\zeta(\rho) \wedge \neg L_{\mathrm{crit}}(\rho)$ projects to non-neutral critical-normalized prime-trace amplitude:

$$A_\zeta(\rho) \wedge \beta \neq \frac{1}{2} \Rightarrow \mathrm{ord}_{1/2}(\rho) = \beta - \frac{1}{2} \neq 0 \Rightarrow x^{\beta-1/2} \neq x^0.$$

The proof explicitly does *not* use

$$x^{\beta-1/2} \neq x^0 \Rightarrow \perp.$$

Only after endpoint-countercase use is fixed, and only after the structural-to-EF bridge classifies the branch as non-neutral order on the fixed prime-trace endpoint image, does the native-counterforce collapse apply. The native off-line branch is not granted a fifth role:

$$\mathrm{NativeCounterforce}_{\text{RH-EF}}(\rho) \Rightarrow \mathrm{BridgeNeutralized}(\rho) \vee \mathrm{SupportOnly}(\rho) \vee \mathrm{DomainShift}_{\mathrm{EF}}(\rho) \vee D_{\mathrm{amp}}(\rho).$$

Inside the exact-complement off-line subproof, bridge-neutralization contradicts the live non-neutral order, support-only status lacks endpoint counterforce, and fixed-domain evaluation excludes domain/interface shift. Thus the only endpoint-standing native-counterforce branch is $D_{\mathrm{amp}}(\rho)$. Equivalently, the projected role becomes local separator occupation:

$$\mathrm{ProjectionRoleContent}_{\mathrm{EF}}(\neg L_{\mathrm{crit}}(\rho), \rho) \wedge \mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho) \Rightarrow \mathrm{AmpSep}^{\mathrm{loc}}(\rho).$$

Then

$$\mathrm{AmpSep}^{\mathrm{loc}}(\rho) \Rightarrow \mathrm{NonCriticalAmpStatusWork}_R(\rho) \Rightarrow D_{\mathrm{amp}}(\rho).$$

Endpoint use of the same local separator act yields

$$\mathrm{AmpSep}^{\mathrm{loc}}(\rho) \Rightarrow \mathrm{EndpointUse}_R(\mathrm{AmpSep}^{\mathrm{loc}}(\rho)) \Rightarrow N_{\mathrm{amp}}(\rho) \Rightarrow \neg D_{\mathrm{amp}}(\rho).$$

Thus $D_{\mathrm{amp}}(\rho) \wedge \neg D_{\mathrm{amp}}(\rho)$, contradiction. The annotation-discharge theorem discharges the original local assumption $[\neg L_{\mathrm{crit}}(\rho)]_i$, not a strengthened conjunction. Therefore $L_{\mathrm{crit}}(\rho)$. Since ρ was arbitrary under $A_\zeta(\rho)$,

$$\forall \rho\, (A_\zeta(\rho) \Rightarrow L_{\mathrm{crit}}(\rho)).$$

This is the classical Riemann Hypothesis.

Anti-trivialization lock. The manuscript does not license the schema $\neg P \Rightarrow \mathrm{Separator}(P) \Rightarrow P$. Endpoint exclusion is permitted only after target instantiation, official route fixation, projection-role content, endpoint-counterforce use, route-locus exhaustion, and the proof that the projected role performs independent same-domain endpoint-status work.

Counterexample-force lock. A counterexample object is not automatically endpoint use. This manuscript distinguishes raw counterexample content, ordinary counterexample theoremhood, official counterexample force, and local exact-complement countercase force. Only official counterexample force and local endpoint countercase force enter endpoint-use discipline. Thus the proof does not classify the object ρ as endpoint use; it classifies the mathematical act of offering ρ, or a theorem asserting its existence, as endpoint-defeating counterexample force against the official RH endpoint.

No one-step retyping. The proof does not infer $D_{\text{amp}}(\rho)$ directly from $\neg L_{\text{crit}}(\rho)$, from $\beta \neq 1/2$, or from $x^{\beta-1/2} \neq x^0$. The off-line branch first appears as ordinary exact-complement content. It becomes endpoint-standing amplitude separator occupation only after the fixed zeta/prime-trace carrier, official explicit-formula route, same-zero projection, critical normalization, projection-role content, and endpoint-counterforce use are in force. This is the safeguard against collapsing native off-line content into its own discriminator.

No-autonomous-negative-endpoint lock. The proof's final same-mode commitment is that ordinary negative endpoint force is not a free-standing fifth role. If the negative branch has no endpoint force, it is raw content, support, bookkeeping, or ordinary theoremhood. If it has endpoint force against the official target on the same carrier, it is endpoint use and is governed by the same kernel-forced consequence layer as any official endpoint act.

1 Counterexample Force as Endpoint Use

This section supplies the final general bridge needed by the native-counterforce layer. A classical reader may agree that the off-line branch projects to non-neutral explicit-formula order and still object that a counterexample is simply a mathematical object satisfying the negation of the endpoint, not endpoint governance. The response is a role distinction. A bare object or proposition is not an endpoint act. The mathematical use of that object or proposition to defeat, displace, or settle the official endpoint on the same fixed carrier is an endpoint act.

The section is deliberately placed before target instantiation. Later RH-specific constructions rely on the following general discipline:

$$\mathrm{OfficialCounterexampleForce}_R(B) \Rightarrow \mathrm{EndpointUse}_R(B).$$

It is the counterexample analogue of ordinary official endpoint resolution. Ordinary theoremhood is allowed. Raw counterexample syntax is allowed. Local reductio assumptions are allowed. The controlled object is endpoint-defeating counterexample force.

1.1 Bare content, theoremhood, official force, and local force

Definition 1.1 (Bare counterexample content)**.** *For a fixed endpoint context R and positive endpoint branch P_R,* $\mathrm{BareCounterexampleContent}_R(B)$ *means that B is raw logical counterexample content whose truth would contradict or refute the positive endpoint proposition on the relevant carrier. It records proposition-level content only. It does not assert proof, reportability, endpoint standing, downstream reuse, official resolution, or endpoint use.*

For RH-EF, the local bare counterexample content attached to a point ρ is

$$\mathrm{BareCounterexampleContent}_{\text{RH-EF}}(\rho) := A_\zeta(\rho) \wedge \neg L_{\mathrm{crit}}(\rho).$$

The global bare negative branch is

$$\mathrm{RH}^-_{\mathrm{bare}} := \exists\rho\, \mathrm{BareCounterexampleContent}_{\text{RH-EF}}(\rho).$$

Definition 1.2 (Ordinary counterexample theoremhood)**.** $\mathrm{OrdCounterexampleThm}_R(B)$ *means ordinary theoremhood of the counterexample branch B:*

$$\mathrm{OrdCounterexampleThm}_R(B) := \mathrm{Proof}_R(B).$$

For RH-EF, this includes a hypothetical ordinary proof of

$$\exists\rho \left(A_\zeta(\rho) \wedge \neg L_{\mathrm{crit}}(\rho)\right).$$

Ordinary counterexample theoremhood is proof legitimacy. It is not, by itself, the independent amplitude discriminator excluded later.

Definition 1.3 (Official counterexample force)**.** $\mathrm{OfficialCounterexampleForce}_R(B)$ *means that a counterexample branch B, or a theorem-bearing act asserting B, is used to defeat, displace, or settle the official endpoint on the same fixed carrier. It includes the following endpoint-use features:*

1. *the official target is fixed;*
2. *the carrier of evaluation is fixed;*

3. *the branch being used as counterexample is fixed;*
4. *the act is theorem-bearing, or is locally assumption-bearing inside an exact-complement subproof;*
5. *the positive endpoint branch is excluded or blocked by the counterexample use;*
6. *the counterexample is reportable as endpoint-defeating content if the act succeeds;*
7. *the counterexample licenses downstream reuse as negative endpoint material if accepted;*
8. *the act is final at act-time rather than repaired into standing later as the same act;*
9. *lawful redescription preserves target, carrier, branch, and endpoint-defeating role.*

Definition 1.4 (Local counterexample force). *For a proof target P and a branch B, write*

$$\mathrm{LocalCounterexampleForce}_R(B; P).$$

This means that B has been opened inside a reductio or proof-by-cases subproof as the live exact-complement countercase whose survival would block P on the fixed endpoint carrier. It is local endpoint-facing countercase use. It is not global official negative endpoint resolution and not theoremhood of B.

For RH-EF, the local form is

$$\mathrm{LocalCounterexampleForce}_{\text{RH-EF}}(\neg L_{\text{crit}}(\rho); L_{\text{crit}}(\rho)).$$

Definition 1.5 (Local endpoint use). $\mathrm{LocalEndpointUse}_R(B)$ *means endpoint use inside a governed local subproof rather than a global endpoint outcome. It fixes the same target, carrier, branch role, local report force, act-time status, and admissible continuation inside the subproof. Local endpoint use is enough to trigger local fixed-domain consequences. It does not assert global official negative resolution.*

1.2 Bare content is not endpoint use

Theorem 1.1 (Bare counterexample content is not endpoint use). *Bare counterexample content alone does not imply endpoint use:*

$$\mathrm{BareCounterexampleContent}_R(B) \not\Rightarrow \mathrm{EndpointUse}_R(B).$$

In the RH-EF instance,

$$A_\zeta(\rho) \wedge \neg L_{\text{crit}}(\rho) \not\Rightarrow \mathrm{EndpointUse}_R(\neg L_{\text{crit}}(\rho)).$$

Proof. Bare content is a proposition-level role. It may be true, false, hypothetical, descriptive, support-level, unproved, or later proved. None of those statuses by itself fixes official endpoint target, reportability, downstream reuse, branch exclusion, act-time finality, or endpoint-role stability. Endpoint use requires an act that offers the branch as endpoint-defeating or endpoint-resolving. Therefore bare counterexample content is not endpoint use. □

Audit 1.1 (Objects are not acts). *The manuscript does not classify the object ρ as endpoint use. It classifies a mathematical act: using ρ, or a theorem asserting its existence, as counterexample force against the official endpoint. An object can be raw content. A counterexample act has endpoint role.*

Theorem 1.2 (Ordinary counterexample theoremhood is kernel-internal, not automatically forbidden)**.** *For any fixed endpoint context,*

$$\mathrm{OrdCounterexampleThm}_R(B) \Rightarrow K_R.$$

But ordinary counterexample theoremhood, considered only as theoremhood, is not the endpoint discriminator later excluded:

$$\mathrm{OrdCounterexampleThm}_R(B) \not\Rightarrow D(B).$$

Proof. A genuine proof is a governed derivation and therefore kernel-internal. Its polarity does not make it defective. The endpoint discriminator arises only when the counterexample theorem is used as official endpoint counterforce or endpoint resolution on the fixed carrier and thereby determines branch standing, reportability, downstream reuse, or exclusion of the positive branch. Ordinary theoremhood alone does not perform that role. □

Audit 1.2 (Negative theoremhood is protected)**.** *A hypothetical ordinary proof of an off-line zero is not rejected merely because it is negative. The paper distinguishes proof legitimacy from endpoint-role occupation. The rejected object, if the endpoint route is fixed, is endpoint-standing amplitude separator authority after official counterexample use.*

1.3 Official counterexample force is endpoint use

Theorem 1.3 (Official counterexample force is endpoint use)**.** *For every fixed endpoint context R and counterexample branch B,*

$$\mathrm{OfficialCounterexampleForce}_R(B) \Rightarrow \mathrm{EndpointUse}_R(B).$$

Proof. A counterexample used only as raw syntax or background possibility is not official counterexample force. By definition, official counterexample force uses the branch to defeat or settle the fixed endpoint on the same carrier. Such an act fixes the target being defeated, fixes the carrier on which defeat is claimed, fixes the negative branch as counterexample branch, fixes the exclusion of the positive branch, and fixes reportability and downstream reuse if the act succeeds. It must be final at act-time, since a later bridge or repair would be a new endpoint act rather than the same counterexample already standing. It is stable under lawful redescription, since otherwise the alleged counterexample would not be a counterexample to the same endpoint. These are exactly the features of theorem-bearing endpoint use. Therefore $\mathrm{OfficialCounterexampleForce}_R(B) \Rightarrow \mathrm{EndpointUse}_R(B)$. □

Corollary 1.1 (No autonomous counterexample endpoint role)**.** *There is no same-carrier official counterexample force that stands outside endpoint-use discipline:*

$$\neg\exists B\,\big(\mathrm{OfficialCounterexampleForce}_R(B) \wedge \neg\mathrm{EndpointUse}_R(B)\big).$$

Equivalently,

$$\neg\mathrm{EndpointUse}_R(B) \Rightarrow \neg\mathrm{OfficialCounterexampleForce}_R(B).$$

Proof. The first display is the direct negation of the possibility excluded by Theorem 1.3. That theorem says: whenever a branch has official counterexample force, it has endpoint-use status. Hence no branch can have official counterexample force while lacking endpoint-use status. The second display is the contrapositive. □

Theorem 1.4 (Autonomous ordinary-negative endpoint role collapse)**.** *Let B be a negative or counterexample branch on a fixed endpoint carrier. If B is not endpoint use, then it cannot defeat the official endpoint as the settled same-carrier counterexample branch. Its remaining lawful roles are raw content, ordinary theoremhood considered only as proof legitimacy, support, bookkeeping, local syntax without endpoint force, or carrier/domain-shifted material.*

Proof. Suppose B is not endpoint use. By Corollary 1.1, B has no official counterexample force. Therefore B cannot simultaneously exclude the positive endpoint branch, fix negative endpoint reportability, license downstream reuse as the settled negative endpoint, and preserve the same carrier and act-time finality. If the branch is present at all, it remains in a role that does not settle the endpoint: raw syntax, proof legitimacy without official endpoint use, support, bookkeeping, local assumption without endpoint force, or a shifted carrier/domain. These cases exhaust the ordinary negative roles that do not exercise official endpoint-defeating force. □

Audit 1.3 (No autonomous ordinary-negative endpoint role)**.** *This is the explicit form of the manuscript's central nonclassical commitment. The proof does not deny ordinary negative content. It denies that ordinary negative content can retain official endpoint-defeating force while refusing endpoint-use status. A same-mode objection must therefore exhibit a counterexample act that defeats the official endpoint while failing to fix the endpoint-use features listed in Definition 2.3.*

Remark 1.1 (Counterexample use, not counterexample object)**.** *The theorem does not say that every object satisfying a negative predicate is an endpoint act. It says that the act of offering such an object, or a theorem asserting its existence, as same-carrier official counterexample force is endpoint use. This is the same role distinction used throughout the manuscript between support-level content and endpoint occupation.*

Theorem 1.5 (Official counterexample force satisfies target adequacy)**.** *If official counterexample force is exercised on a fixed domain, then the kernel and A+ consequence layer are in force:*

$$\mathrm{OfficialCounterexampleForce}_R(B) \wedge \mathrm{FixedDomain}(R) \Rightarrow K_R \wedge A_R^+(B).$$

Proof. By Theorem 1.3, official counterexample force is endpoint use. Endpoint use is governed construction: it fixes target, carrier, branch, theorem-status or local countercase force, downstream reuse, act-time finality, and redescription-stable role. The target-adequacy-to-kernel theorem gives K_R. Fixed-domain endpoint use then triggers the A+ consequence package: AMetric no-selector discipline, unique admissible interior, report preservation, no carrier transfer, no same-act repair, no third endpoint status, and no independent same-domain endpoint discriminator. Hence $K_R \wedge A_R^+(B)$. □

Audit 1.4 (Why counterexample force is governed)**.** *A counterexample with no fixed target is not a counterexample to this endpoint. A counterexample with no fixed carrier may be a different problem. A counterexample with no reportability or downstream reuse is support or syntax. A counterexample whose standing is supplied only later is a repaired act. Thus a same-carrier official counterexample act has already accepted the endpoint-use features whose consequences the proof applies.*

1.4 Local exact-complement counterexample force

Theorem 1.6 (Exact-complement local force is local endpoint use)**.** *In a fixed endpoint-closure proof, if B is opened as the live exact-complement countercase to target P and is used with endpoint counterforce, then it is local endpoint use:*

$$\mathrm{LocalCounterexampleForce}_R(B;P) \wedge \mathrm{EndpointCounterforce}_R(B) \Rightarrow \mathrm{LocalEndpointUse}_R(B).$$

For the RH-EF local subproof,

$$\begin{aligned}&\mathrm{LocalCounterexampleForce}_{\text{RH-EF}}(\neg L_{\mathrm{crit}}(\rho); L_{\mathrm{crit}}(\rho))\\&\quad\wedge \mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho)\\&\quad\Rightarrow \mathrm{LocalEndpointUse}_R(\neg L_{\mathrm{crit}}(\rho)).\end{aligned}$$

Proof. Local counterexample force is the local proof role in which the exact complement is introduced to block the endpoint target. Endpoint counterforce says that the branch is not inert syntax but is being used to defeat the target inside the subproof. Together they fix the local target, the local branch role, the same carrier, the exclusion relation to the positive target, and the act-time subproof status. Those are the local analogues of endpoint use. The conclusion is local endpoint use, not global negative endpoint outcome. □

Theorem 1.7 (Local endpoint use triggers local A+ closure)**.** *On a fixed domain,*

$$\mathrm{LocalEndpointUse}_R(B) \wedge \mathrm{FixedDomain}(R) \Rightarrow A^+_{R,\mathrm{loc}}(B).$$

For the RH-EF amplitude separator case, local A+ includes the no-independent-amplitude-discriminator closure:

$$A^+_{R,\mathrm{loc}}(B) \Rightarrow N_{\mathrm{amp}}(\rho).$$

Proof. The kernel roles do not suspend inside a reductio subproof. A local endpoint act still has target, carrier, branch role, standing-bearing local force, admissible continuation, and act-time status. Therefore fixed-domain closure supplies the local A+ consequences. In the RH-EF setting, the relevant local consequence is the no-independent-amplitude-discriminator closure $N_{\mathrm{amp}}(\rho)$. □

Audit 1.5 (No global promotion)**.** *The local A+ theorem does not promote the local assumption to global official negative resolution. It only states that a governed local endpoint countercase on the same fixed carrier cannot use a local independent discriminator while the kernel-forced consequence layer is in force.*

1.5 RH-EF specialization

Definition 1.6 (RH off-line counterexample force)**.** RHOffLineCounterexampleForce(ρ) *means that the bare analytic content*

$$A_\zeta(\rho) \wedge \neg L_{\mathrm{crit}}(\rho)$$

is used to defeat the official Riemann Hypothesis endpoint on the fixed zeta/explicit-formula carrier. It is stronger than off-line coordinate description and stronger than support-level amplitude displacement.

Theorem 1.8 (RH off-line counterexample force is endpoint use)**.**

$$\mathrm{RHOffLineCounterexampleForce}(\rho) \Rightarrow \mathrm{EndpointUse}_R(\mathrm{RH}^-_{\mathrm{bare}}).$$

Proof. RH off-line counterexample force is the RH-EF instance of official counterexample force. It uses the off-line zerohood branch to defeat the official RH endpoint on the same fixed carrier. The general counterexample-force endpoint-use theorem therefore applies. □

Theorem 1.9 (RH off-line counterexample force projects to EF role content)**.** *Under the official explicit-formula route,*

$$\begin{aligned}&\mathrm{RHOffLineCounterexampleForce}(\rho) \wedge \mathrm{OfficialEFRoute}(R,\zeta,\psi)\\&\quad\Rightarrow \mathrm{ProjectionRoleContent}_{\mathrm{EF}}(\neg L_{\mathrm{crit}}(\rho),\rho).\end{aligned}$$

Proof. The official EF route binds branch roles to explicit-formula amplitude roles. Off-line counterexample force is not merely coordinate description; it is the endpoint-defeating use of the off-line branch. Under projection, that branch has non-neutral EF order as its endpoint-role content. This is precisely $\mathrm{ProjectionRoleContent}_{\mathrm{EF}}(\neg L_{\mathrm{crit}}(\rho), \rho)$. □

Theorem 1.10 (RH off-line counterexample force enters native-counterforce exhaustion)**.**

$$\begin{aligned}
&\mathrm{RHOffLineCounterexampleForce}(\rho) \wedge \mathrm{OfficialEFRoute}(R, \zeta, \psi) \\
&\quad \wedge \mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R) \\
&\quad \Rightarrow \mathrm{NativeCounterforce}_{\mathrm{RH\text{-}EF}}(\rho).
\end{aligned}$$

Proof. By the preceding theorem, the off-line counterexample act has projected EF role content. The model adequacy theorem identifies that content with non-neutral critical-normalized order for the same zerohood instance. Since the use is endpoint-defeating counterexample force rather than descriptive or support-only use, it is native endpoint counterforce in the sense used by the native-counterforce collapse theorem. □

Corollary 1.2 (No separate ordinary-counterexample escape)**.** *For RH-EF, official off-line counterexample force has no endpoint role outside the native-counterforce exhaustion:*

$$\begin{aligned}
\mathrm{RHOffLineCounterexampleForce}(\rho) \Rightarrow \mathrm{BridgeNeutralized}(\rho) \vee \mathrm{SupportOnly}(\rho) \vee \mathrm{Bookkeeping}(\rho) \\
\vee \mathrm{DomainShift}_{\mathrm{EF}}(\rho) \vee D_{\mathrm{amp}}(\rho).
\end{aligned}$$

Inside the live off-line endpoint countercase on a fixed domain, the first four branches are unavailable, so $D_{\mathrm{amp}}(\rho)$ *remains.*

Proof. The first statement follows from entry into native-counterforce exhaustion. In the live local off-line subproof, bridge-neutralization gives endpoint-neutral order and hence conflicts with $\neg L_{\mathrm{crit}}(\rho)$; support-only and bookkeeping lack endpoint counterforce; fixed-domain endpoint evaluation excludes domain/interface shift. Therefore the remaining exhaustive endpoint role is $D_{\mathrm{amp}}(\rho)$. □

Audit 1.6 (Where the classical objection now lands)**.** *After this section, a classical objection cannot merely say that a counterexample is an object rather than governance. The manuscript agrees. The object is raw content. The endpoint-defeating use of that object is an endpoint act. A same-mode denial must now show a same-carrier official counterexample act that defeats the endpoint while failing to fix target, carrier, branch, reportability, downstream reuse, branch exclusion, act-time finality, or redescription-stable role.*

2 Official Endpoint, Target Instantiation, and Context Non-Smuggling

2.1 Official target and branch split

Definition 2.1 (Nontrivial zerohood and critical-line readout)**.** *For a complex number ρ, define*

$$A_\zeta(\rho) := \big(\zeta(\rho) = 0\big) \wedge \big(0 < \mathrm{Re}(\rho) < 1\big),$$

and

$$L_{\mathrm{crit}}(\rho) := \big(\mathrm{Re}(\rho) = 1/2\big).$$

The official positive endpoint is

$$\mathrm{RH}^+ := \forall\rho\,(A_\zeta(\rho) \Rightarrow L_{\mathrm{crit}}(\rho)).$$

The bare negative branch is

$$\mathrm{RH}^-_{\mathrm{bare}} := \exists\rho\,\big(A_\zeta(\rho) \wedge \neg L_{\mathrm{crit}}(\rho)\big).$$

Lemma 2.1 (Raw complementarity)**.** *At the bare analytic proposition layer,*

$$\neg\mathrm{RH}^+ \iff \mathrm{RH}^-_{\mathrm{bare}}.$$

Locally, under $A_\zeta(\rho)$, the exact complement of $L_{\mathrm{crit}}(\rho)$ is $\neg L_{\mathrm{crit}}(\rho)$, equivalently $\mathrm{Re}(\rho) \neq 1/2$.

Proof. The first equivalence is classical quantifier negation applied to $\forall\rho(A_\zeta(\rho) \Rightarrow L_{\mathrm{crit}}(\rho))$. The local equivalence follows from the definition of L_{crit}. No endpoint standing, reportability, or theorem status is inferred from this bare complementarity. □

Audit 2.1 (Raw complementarity is not endpoint standing)**.** *The proof uses raw complementarity only to open a local reductio branch and later discharge it. It does not infer that the negative branch has official endpoint standing. The standing-bearing role is supplied only by exact-complement countercase use in the fixed endpoint-closure proof, and then only locally.*

2.2 The RH-EF context object

Definition 2.2 (Fixed RH explicit-formula endpoint context)**.** *The object*

$$C_{\text{RH-EF}}(R, \zeta, \psi)$$

records the official RH endpoint context through the explicit-formula route. It fixes the zeta carrier, the prime-trace carrier, nontrivial zerohood, critical-line readout, the positive endpoint branch, the bare negative branch, the critical-normalized amplitude image, the local countercase slot, the endpoint counterexample slot, and the amplitude separator occupation slot.

Theorem 2.1 (Context non-smuggling)**.** *The context object $C_{\text{RH-EF}}(R, \zeta, \psi)$ does not assert any of the following:*

$$\mathrm{RH}^+, \quad L_{\mathrm{crit}}(\rho), \quad \neg\mathrm{RH}^-_{\mathrm{bare}}, \quad A^+_R(B),$$
$$N_{\mathrm{amp}}(\rho), \quad \neg D_{\mathrm{amp}}(\rho), \quad \neg\mathrm{AmpSep}^{\mathrm{loc}}(\rho), \quad \mathrm{RHOffLineCounterexampleForce}(\rho).$$

It fixes carrier, route, branch slots, endpoint-counterexample slots, and endpoint-image roles only.

Proof. By definition the context object records what is being evaluated and by which route. The zeta carrier does not include line support; the prime-trace carrier does not assert neutral amplitude occupation for all zeros; the route predicate does not assert the endpoint. All status consequences are derived later from endpoint use, model adequacy, projection, and the kernel-forced consequence layer. ☐

Audit 2.2 (Context smuggling objection). *A hostile objection must identify a clause inside $C_{\text{RH-EF}}$ that already states line membership, neutral amplitude occupation for all zeros, no-independent closure, or exclusion of the off-line branch. Merely fixing the critical-line endpoint image is not smuggling; it is target instantiation. The positive image is fixed, not asserted as universally occupied.*

2.3 Official explicit-formula endpoint route

Definition 2.3 (Official explicit-formula route). $\mathrm{OfficialEFRoute}(R, \zeta, \psi)$ *means that the official RH endpoint is being evaluated through the standard explicit-formula relation between zeta zerohood and prime-trace oscillation, while preserving the same zeta zerohood carrier, the same prime-trace carrier, and the same critical-line readout target. It is stronger than mentioning the explicit formula as background evidence.*

Definition 2.4 (Endpoint-closure evaluation). $\mathrm{OSEval}_{\text{RH-EF}}(R, \zeta, \psi)$ *means that the official RH endpoint is being evaluated in the endpoint-closure proof class on the fixed explicit-formula carrier. It fixes target, carrier, route, branch slots, and local counterforce discipline. It is not endpoint-complete evaluation and does not assert either branch.*

Theorem 2.2 (Official route is target instantiation, not endpoint assertion).

$$C_{\text{RH-EF}}(R, \zeta, \psi) \wedge \mathrm{OfficialEFRoute}(R, \zeta, \psi) \not\Rightarrow \mathrm{RH}^{+}.$$

The route supplies the bridge by which branch roles are read; it does not supply line support.

Proof. The official route says how endpoint roles are projected onto the prime-trace amplitude image. Projection is not occupation. In particular, a route can map the positive branch to neutral amplitude and the negative branch to non-neutral amplitude without deciding which branch is true. Therefore no positive endpoint follows from route fixation alone. ☐

3 Kernel, A+, ATS, and UEAP Replay

3.1 Kernel-neutral target adequacy

Definition 3.1 (Kernel-neutral target adequacy). *A theorem-bearing endpoint act is target-adequate when it satisfies: target determinacy, step evaluability, act-time finality, and same-regime fidelity. These conditions are neutral with respect to AASC vocabulary; they are the conditions under which a raw trace can be assessed as a determinate same-carrier mathematical act.*

Theorem 3.1 (Target adequacy forces the kernel). *If an endpoint act is target-adequate, then reference, standing, admissibility, and irreversibility are already in force for that act:*

$$TargetAdequacy_R(B) \Rightarrow Ref_R \wedge St_R \wedge Adm_R \wedge Irr_R.$$

Proof. Target determinacy supplies fixed reference. Step evaluability supplies standing-bearing step status. Act-time finality supplies irreversibility: later repair is a new act, not the same act made valid. Same-regime fidelity supplies the admissibility boundary preserving those statuses under lawful redescription. Thus the kernel roles are forced roles, not optional AASC additions. □

Theorem 3.2 (Endpoint use is governed construction)**.** *For any fixed endpoint context R and branch or local endpoint act B,*

$$\mathrm{EndpointUse}_R(B) \Rightarrow GovernedConstruction_R(B) \Rightarrow K_R.$$

Proof. Endpoint use fixes target, carrier, theorem-status or countercase status, downstream reuse, act-time finality, and redescription-stable role. These are exactly the features of governed construction on a fixed domain. The kernel-forcing theorem applies. □

3.2 A+ consequence package

Definition 3.2 (A+ for the RH-EF endpoint)**.** *$A_R^+(B)$ is the fixed-domain consequence layer forced by kernel-instantiated endpoint use. For the RH-EF route it includes boundary fixation, endpoint bivalence, AMetric no-selector discipline, standing as reuse-stable admissibility, conservation of standing, unique admissible interior, no raw trace sufficiency, no carrier transfer, no post-selection repair, no silent redescription, no third endpoint status, report preservation, ATS anchor/tensor/skin stability, and no independent same-domain endpoint-status discriminator.*

Theorem 3.3 (Kernel forces A+)**.**

$$\mathrm{EndpointUse}_R(B) \wedge \mathrm{FixedDomain}(R) \Rightarrow A_R^+(B).$$

Proof. By endpoint use the kernel is in force. Fixed-domain closure applies the kernel consequences: bivalence and AMetricity at the boundary, uniqueness of the admissible interior, standing conservation, no raw-trace sufficiency, no carrier transfer, no same-act repair, no skin authority, and no independent same-domain discriminator. A+ is therefore a consequence layer rather than an independent premise. □

Audit 3.1 (A+ is not doing analytic estimation)**.** *A+ does not assert a numerical bound for $\psi(x) - x$, does not prove a zero-free region, and does not alter the explicit formula. It controls whether a theorem-bearing endpoint act may introduce a second same-domain endpoint-status classifier. The analytic transport to amplitude status is supplied separately by the explicit-formula route.*

3.3 ATS local packet

Definition 3.3 (Local ATS packet)**.** $\mathrm{ATS}_{\text{RH-EF}}$ *is the conjunction of: fixed zeta anchor, fixed prime-trace anchor, same-zero identity, fixed critical normalization, explicit-formula route preservation, tensor status of the exponent $\beta - 1/2$, skin-inertness of presentation choices, and no second same-domain amplitude classifier.*

Theorem 3.4 (ATS blocks skin authority)**.** *Under* $\mathrm{ATS}_{\text{RH-EF}}$*, a change of notation, smoothing presentation, truncation convention, or equivalent formula display does not create endpoint standing. If it changes the same-zero contribution, the normalized exponent, or the prime-trace carrier, it is not skin but a carrier or tensor change requiring bridge discharge.*

Proof. ATS separates anchors from tensor content and from skin. Endpoint status can be carried only by anchor-preserving tensor content. A skin change carries no standing. A non-skin change alters the target or carrier and therefore cannot be used as same-domain endpoint resolution without fresh bridge certification. □

3.4 UEAP local packet

Definition 3.4 (UEAP report ladder). *The RH-EF report ladder is:*

1. *zerohood report* $A_\zeta(\rho)$*;*
2. *coordinate report* $\beta \neq 1/2$*;*
3. *explicit-formula contribution report* x^ρ/ρ*;*
4. *support-level amplitude report* $x^{\beta-1/2} \neq x^0$*;*
5. *endpoint-counterforce report;*
6. *endpoint-result report.*

Theorem 3.5 (UEAP no-overreporting for amplitude). *Support-level non-neutral amplitude does not by itself become endpoint counterforce:*

$$OffLineAmpSupport(\rho) \not\Rightarrow \mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho).$$

Endpoint counterforce requires official route fixation, local exact-complement use, and role occupation.

Proof. UEAP forbids a report from exceeding its carrier, bridge, and support. The amplitude report says what scale is contributed by a zero under the explicit formula. It does not by itself assert endpoint resolution. Endpoint counterforce is a higher report role: the amplitude displacement is used to defeat the critical-neutral endpoint image on the same carrier. That use must be separately established. □

3.5 Official endpoint resolution target adequacy

Theorem 3.6 (Official RH-EF endpoint resolution satisfies target adequacy). *For any branch or local countercase act B,*

$$\mathrm{OrdOfficialRes}_R(B) \wedge \mathrm{OfficialEFRoute}(R, \zeta, \psi) \wedge \mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R) \wedge \mathrm{FixedDomain}(R) \Rightarrow K_R \wedge A_R^+(B).$$

Proof. Target determinacy holds because the endpoint is the official RH assertion, the carrier is the fixed zeta/prime-trace carrier, the route is the explicit formula, and the endpoint image is critical-normalized amplitude. Step evaluability holds because theorem-bearing use classifies a branch as endpoint outcome, support, bookkeeping, or counterforce. Act-time finality holds because later analytic certification creates a new act rather than repairing an earlier unbridged endpoint report. Same-regime fidelity holds because lawful redescription preserves the zeta carrier, prime-trace carrier, same-zero identity, normalization, and branch role. Target adequacy forces the kernel, and the kernel on a fixed domain forces A+. □

3.6 Local RH-EF kernel packet K1–K13

The preceding replay is now recorded in local endpoint vocabulary. This packet is not an additional analytic premise. It states the fixed-domain consequences already forced when the official RH endpoint is evaluated through the explicit-formula amplitude carrier.

Node	General necessity	RH-EF form
K1	Fixed-domain kernel	A theorem-bearing RH-EF endpoint act fixes the zeta carrier, nontrivial zerohood, same-zero identity, explicit-formula route, prime-trace carrier, critical normalization, branch role, report status, and act-time finality.
K2	Reference fixation	A report about ρ cannot silently replace the zero, the zeta carrier, the explicit-formula contribution, the normalization, the prime-trace image, or the endpoint role.
K3	Standing-readout separation	$A_\zeta(\rho)$ is zerohood standing; $L_{\mathrm{crit}}(\rho)$ is critical-line readout; $\mathrm{ord}_{1/2}(\rho) = 0$ is explicit-formula neutral-order readout.
K4	Bivalence and fail-closed status	Unknown, open, finite-verified, numerically supported, smoothed, averaged, spectral-model, density-supported, or cancellation-supported status is not a third RH endpoint truth branch.
K5	Unique admissible interior	The fixed critical-neutral amplitude endpoint slot cannot carry plural independent endpoint-governing interiors.
K6	Standing-admissibility identity	Endpoint-standing non-neutral amplitude cannot float outside admissibility; if it has endpoint force, it is typed by the RH-EF endpoint route.
K7	No generators	Writing $A_\zeta(\rho) \wedge \neg L_{\mathrm{crit}}(\rho)$ or $\mathrm{ord}_{1/2}(\rho) \neq 0$ does not generate endpoint authority for itself.
K8	No carriers	Spectral models, Hilbert–Polya candidates, random-matrix analogues, finite verification lists, positivity criteria, density estimates, smoothing regimes, or surrogate prime traces cannot carry RH endpoint status without a lawful bridge back to the fixed RH-EF carrier.
K9	No post-selection repair	Later cancellation, smoothing refinement, zero-free region, density theorem, spectral construction, or positivity certificate is a new bridge-complete act, not repair of an earlier unbridged endpoint report.
K10	AMetric no-selector boundary	No preferred zero representative, pairing convention, smoothing parameter, truncation height, finite verification bound, spectral model, or normalization variant decides endpoint status before admissibility is fixed.
K11	UEAP report preservation	A report may not exceed its zerohood, contribution, normalization, bridge, and support. Coordinate or amplitude support cannot be reported as endpoint result without endpoint-use role.

K12	ATS skin discipline	Notation, smoothing display, truncation convention, pairing display, finite-height language, or model analogy cannot become endpoint tensor authority by presentation alone.
K13	Endpoint exhaustion	Once carrier, zerohood, EF route, critical-normalized endpoint image, and admissibility boundary are fixed, endpoint-bearing branches are exhausted by neutral amplitude occupation or governed non-neutral endpoint use. Support, bookkeeping, carrier shift, repair, open status, and hidden-fifth statuses do not occupy endpoint truth.

Theorem 3.7 (Local RH-EF kernel packet K1–K13)**.** *For every official RH-EF endpoint use, the local consequences K1–K13 in the displayed packet are in force on the fixed zeta/prime-trace endpoint carrier.*

Proof. Endpoint use fixes target, branch, carrier, theorem-status or local countercase status, downstream reuse, act-time finality, and lawful-redescription stability. By the kernel-forcing theorems above, reference, standing, admissibility, and irreversibility are already active. K1 records their joint fixed-domain role. K2 and K3 are reference and role-separation consequences. K4–K6 are the bivalent, unique-interior, and standing-admissibility consequences. K7–K10 block self-authorization, carrier import, repair, and selector authority. K11 and K12 record UEAP report preservation and ATS skin discipline. K13 is the endpoint exhaustion consequence specialized to the critical-neutral amplitude image. These clauses are therefore consequences of the endpoint-use regime, not new analytic assumptions. □

4 Explicit-Formula Prime-Trace Model Adequacy

4.1 Prime-trace carrier and critical normalization

Let

$$\psi(x) = \sum_{n \le x} \Lambda(n), \qquad E_\psi(x) = \psi(x) - x,$$

and

$$\mathcal{E}_\psi(x) = \frac{\psi(x) - x}{x^{1/2}}.$$

The normalization by $x^{1/2}$ is not an assertion of RH. It is the critical-line endpoint normalization: on the positive endpoint image, a zero contribution has neutral magnitude exponent.

Definition 4.1 (Explicit-formula contribution role)**.** *For a nontrivial zero $\rho = \beta + i\gamma$, the explicit-formula contribution role is represented, suppressing smoothing and harmless presentation choices, by a term of scale*

$$\frac{x^\rho}{\rho}.$$

After critical normalization, the amplitude profile is governed by

$$\frac{x^{\rho-1/2}}{\rho},$$

and its magnitude exponent is $\beta - 1/2$.

Definition 4.2 (Context-bound explicit-formula model adequacy). $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$ *is the adequate explicit-formula endpoint model when the following clauses hold:*

(EF1) zeta-zero coverage: every nontrivial zerohood instance in the official RH carrier appears in the zero contribution class;

(EF2) contribution soundness: every contribution used by the model is attached to the same zeta carrier;

(EF3) same-zero identity preservation: the contribution attributed to ρ remains tied to the same zerohood instance;

(EF4) symmetry-pair awareness: paired and conjugate contributions are lawful but do not erase individual scale-class typing unless a bridge proves endpoint-neutral cancellation;

(EF5) smoothing/truncation discipline: test functions and truncations are skin or support unless they change endpoint role;

(EF6) amplitude exactness: $\rho = \beta + i\gamma$ has critical-normalized amplitude scale $x^{\beta-1/2}$;

(EF7) critical-neutral exactness: $\beta = 1/2$ iff the normalized amplitude exponent is zero;

(EF8) off-line displacement exactness: $\beta \neq 1/2$ iff the normalized amplitude exponent is nonzero;

(EF9) no RH smuggling: no clause asserts that all zero contributions are neutral;

(EF10) support/endpoint separation: amplitude data are lawful support unless endpoint counterforce use is supplied.

Theorem 4.1 (Model adequacy).

$$C_{\text{RH-EF}}(R, \zeta, \psi) \wedge \mathrm{FixedDomain}(R) \wedge \mathrm{OfficialEFRoute}(R, \zeta, \psi) \Rightarrow \mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R).$$

Proof. The context fixes the official zeta and prime-trace carriers. The route fixes the explicit formula as the zero-to-prime-trace projection. The zero contribution of ρ has the standard scale x^{ρ}/ρ; critical normalization divides by $x^{1/2}$, leaving exponent $\beta - 1/2$. Symmetry, smoothing, truncation, and equivalent formula presentations are included as lawful carrier data but do not assert universal neutral occupation. Therefore the adequacy clauses hold. □

Audit 4.1 (Model adequacy burden). *A valid model-adequacy objection must identify a nontrivial zero lost by the contribution model, a contribution not tied to the zeta carrier, a failure of same-zero identity, a normalization mismatch, a hidden RH assumption, or a support/endpoint collapse. Merely objecting that the explicit formula contains error terms or smoothing choices does not meet the burden unless those choices alter endpoint role.*

4.2 Term isolation and pairing closure

Definition 4.3 (Term-isolation discipline). *A zero contribution may be treated as endpoint-relevant amplitude content only when same-zero identity, explicit-formula route, normalization, and contribution class are fixed. Term isolation is not a claim of analytic dominance, absence of cancellation, or global asymptotic lower bound.*

Theorem 4.2 (Term-isolation discipline). *If* $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$ *holds and* $A_{\zeta}(\rho)$*, then the contribution associated with* $\rho = \beta + i\gamma$ *has a well-defined critical-normalized scale class* $\beta - 1/2$*. This scale class may be used in endpoint-role analysis without asserting that the single term dominates the entire prime trace.*

Proof. Model adequacy fixes same-zero identity and the explicit-formula contribution class. The scale class is an algebraic feature of the contribution after normalization. Dominance and cancellation are separate analytic questions. The endpoint analysis requires only that off-line zerohood projects to non-neutral scale class, not that this class dominates all other terms. □

Theorem 4.3 (Pairing closure). *Suppose $\rho = \beta + i\gamma$ is off line. The symmetry-related family of zero contributions is lawful analytic content. However, pairing does not change the individual scale-class fact $\beta - 1/2 \neq 0$ unless a bridge proves that the paired family is endpoint-neutral in the same critical-normalized image. Therefore symmetry pairing is support or bridge material, not automatic neutral occupation.*

Proof. The functional equation and conjugation symmetries generate related zeros and paired terms. Those terms may combine in oscillatory or real-valued ways. But the endpoint image under audit is critical-neutral amplitude occupation for the same zerohood contribution class. Pairing can participate in a proof that the endpoint image is neutral only if it provides a bridge from the paired contribution family to neutral occupation. Without that bridge, it is lawful analytic structure, not erasure of non-neutral scale-class typing. □

Audit 4.2 (Pairing/cancellation non-overclaim). *The theorem does not say off-line contributions dominate the prime trace. It says cancellation does not license endpoint-neutral occupation unless proved as endpoint-neutral bridge content. This blocks the hostile objection that the paper ignores paired sums while avoiding the stronger analytic claim that no cancellation can occur.*

Theorem 4.4 (Pairing/cancellation is not an autonomous native endpoint role). *On the fixed RH-EF carrier, functional-equation pairing, conjugate pairing, smoothing cancellation, aggregate trace cancellation, or any equivalent cancellation presentation has no autonomous endpoint role outside the RH-EF endpoint taxonomy. If such cancellation is used in connection with an off-line contribution, it is exactly one of the following: support-level analytic material; bridge-complete neutralization of the same-zero, same-route, critically normalized endpoint image; a carrier, route, same-zero, normalization, or endpoint-image shift; bookkeeping; or independent same-domain amplitude endpoint-status work. In the last case it is $D_{\text{amp}}(\rho)$.*

Proof. Pairing and cancellation are lawful analytic operations, but their endpoint role is determined by what they do to the fixed endpoint image. If they are used only to organize terms, motivate a bridge, refine a trace formula, or explain analytic structure, they are support-level material. If they prove that the same-zero contribution class is endpoint-neutral in the critical-normalized image, then they are bridge-complete neutralization and yield the positive neutral-order bridge. If they avoid non-neutral order only by changing the zero under audit, the route, the contribution class, the normalization, the prime-trace carrier, or the endpoint image, then they are a carrier or interface shift. If they change no endpoint standing, reportability, reuse, branch exclusion, or admissible continuation, they are bookkeeping. The only remaining case is same-carrier endpoint-status work: cancellation is used to decide whether the non-neutral branch defeats or settles the endpoint while neither supplying neutral bridge completion nor exiting the carrier. That remaining role is independent same-domain amplitude endpoint-status discrimination, namely $D_{\text{amp}}(\rho)$. □

Audit 4.3 (Cancellation-role closure). *A cancellation argument is preserved as analytic mathematics. It becomes endpoint-decisive only by supplying bridge-complete neutralization or by taking an endpoint-status role on the fixed RH-EF carrier. There is no fifth cancellation route that both affects the endpoint and avoids bridge completion, carrier shift, bookkeeping, and $D_{\text{amp}}(\rho)$.*

5 Branch Projection and the Single Amplitude Endpoint Interface

5.1 Positive and negative amplitude projections

Definition 5.1 (Critical-neutral amplitude occupation)**.** *For a fixed zerohood instance $\rho = \beta + i\gamma$, define* $\mathrm{CritAmpOcc}(\rho)$ *by*

$$\mathrm{CritAmpOcc}(\rho) :\Longleftrightarrow x^{\beta-1/2} = x^0$$

as critical-normalized amplitude role occupation. Equivalently, under model adequacy, $\mathrm{CritAmpOcc}(\rho)$ *is the amplitude image of $L_{\mathrm{crit}}(\rho)$.*

Definition 5.2 (Off-line amplitude exclusion)**.**

$$\mathrm{OffLineAmpExcl}(\rho) :\Longleftrightarrow x^{\beta-1/2} \neq x^0.$$

This is non-occupation of the critical-neutral amplitude image. It is support-level analytic content unless used as endpoint counterforce.

Theorem 5.1 (Explicit-formula branch projection)**.** *Under* $\mathrm{OfficialEFRoute}(R, \zeta, \psi) \wedge \mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$, *for each zerohood instance ρ,*

$$L_{\mathrm{crit}}(\rho) \Longleftrightarrow \mathrm{CritAmpOcc}(\rho),$$

and

$$\neg L_{\mathrm{crit}}(\rho) \Longleftrightarrow \mathrm{OffLineAmpExcl}(\rho).$$

Proof. Write $\rho = \beta + i\gamma$. By definition $L_{\mathrm{crit}}(\rho)$ is $\beta = 1/2$. By amplitude exactness, critical-normalized scale has exponent $\beta - 1/2$. Thus $\beta = 1/2$ exactly when the scale is x^0, and $\beta \neq 1/2$ exactly when the scale is non-neutral. This proves both equivalences. □

Audit 5.1 (Projection is not truth selection)**.** *The projection theorem does not decide which branch holds. It tells us what the branch means on the explicit-formula amplitude image if that branch is used in the official route. This mirrors target instantiation: branch roles are fixed before branch truth is decided.*

5.2 Projection-role content

Definition 5.3 (Projection-role content)**.** $\mathrm{ProjectionRoleContent}_{\mathrm{EF}}(B, \rho)$ *means that, under the official explicit-formula route, the endpoint-use act B has the explicit-formula amplitude role associated with ρ as its endpoint-role content. For the local negative branch, this content is* $\mathrm{OffLineAmpExcl}(\rho)$.

Theorem 5.2 (Local off-line countercase binds to projection-role content)**.**

$$\begin{aligned} &A_\zeta(\rho) \wedge \mathrm{OfficialEFRoute}(R, \zeta, \psi) \wedge \mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R) \\ &\quad \wedge \mathrm{ReductioCountercase}_{\text{RH-EF}} \\ &\qquad\qquad (\neg L_{\mathrm{crit}}(\rho); L_{\mathrm{crit}}(\rho)) \\ &\Rightarrow \mathrm{ProjectionRoleContent}_{\mathrm{EF}}(\neg L_{\mathrm{crit}}(\rho), \rho). \end{aligned}$$

Proof. The local assumption is the exact complement of critical-line readout for the fixed zerohood instance. The official route requires that branch content be read through the explicit-formula amplitude image. By branch projection, the complement of line readout is non-neutral amplitude exclusion. The projection-role predicate records that this is the endpoint-role content of the local countercase, not a free-standing descriptive assertion. □

Audit 5.2 (Projection-role content is not yet contradiction)**.** *Projection-role content is still not forbidden. The act becomes separator occupation only when the projected role is used as endpoint counterforce and is not support, bookkeeping, carrier shift, repair, surrogate substitution, or coequal occupation.*

5.3 Single amplitude interface

Theorem 5.3 (Single amplitude endpoint interface)**.** *On the fixed RH-EF carrier, critical-neutral amplitude occupation has exactly one occupation interface:*

$$\mathrm{CritAmpImageOcc}(\rho) \iff \mathrm{CritAmpOcc}(\rho) \iff L_{\mathrm{crit}}(\rho).$$

Thus the positive endpoint image is neutral critical-normalized amplitude occupation, not a family of coequal neutral and non-neutral occupants.

Proof. The endpoint image is the explicit-formula projection of critical-line readout. Critical-line readout is $\beta = 1/2$, which is exactly normalized exponent zero. A nonzero normalized exponent is non-occupation of this image. Treating non-neutral scale as coequal occupation changes from the critical-neutral endpoint image to a broader symmetric outcome carrier. It is not the same endpoint image. □

Corollary 5.1 (Off-line amplitude is non-occupation)**.**

$$\mathrm{OffLineAmpExcl}(\rho) \Rightarrow \neg\mathrm{CritAmpImageOcc}(\rho).$$

Proof. Off-line amplitude exclusion means non-neutral exponent; the single-interface theorem says occupation requires neutral exponent. Hence non-occupation follows. □

Theorem 5.4 (Non-neutral amplitude is not a lawful coequal RH-EF endpoint role)**.** *On the fixed RH-EF endpoint image, critical-neutral amplitude occupation has one interface:*

$$\mathrm{ord}_{1/2}(\rho) = 0.$$

Non-neutral amplitude is lawful as coordinate content, explicit-formula support, local hypothesis, ordinary theoremhood, or broad negative-outcome content. It is not a coequal occupant of the critical-neutral endpoint image. Treating it as coequal occupation changes the interface from critical-neutral amplitude occupation to the symmetric outcome ledger $\{\mathrm{RH}, \neg\mathrm{RH}\}$*, which is an interface reset unless explicitly declared as a different problem carrier.*

Proof. The official RH-EF endpoint image is critical-line readout projected to neutral critical-normalized order. Occupation of that image is $\mathrm{ord}_{1/2}(\rho) = 0$. Non-neutral order $\mathrm{ord}_{1/2}(\rho) \neq 0$ is the exact complement of that occupation interface. It may be recorded as the ordinary negative outcome of a broader problem ledger, but that broader ledger is not the endpoint image whose occupant is neutral amplitude. Moving from the neutral-order image to a symmetric two-outcome ledger changes the endpoint interface. Therefore non-neutral amplitude is non-occupation on the fixed image, not coequal occupation of it. □

Theorem 5.5 (Single-interface non-explosiveness)**.** *The manuscript does not infer*

$$\mathrm{OffLineAmpExcl}(\rho) \Rightarrow \bot.$$

It permits non-neutral amplitude as descriptive analytic consequence, support-level hypothetical content, and ordinary counterexample syntax. Contradiction arises only from endpoint-standing use of that non-neutral amplitude as same-carrier separator occupation.

Proof. The proof later adds endpoint-counterforce use, non-support status, no carrier shift, no repair, no surrogate substitution, and no coequal occupation. Without those additional hypotheses, non-neutral amplitude is a lawful analytic description of what an off-line zero contribution would look like. □

6 Structural-to-Explicit-Formula Equivalence Bridge

This section supplies the load-bearing bridge from the structural RH role-compression paper into the explicit-formula amplitude carrier. Its purpose is to prevent the amplitude proof from relying on a loose analogy. The structural RH paper separates analytic carrier, zerohood standing, critical-line readout, and carrier-preservation bridge. The present manuscript instantiates that bridge through the explicit-formula amplitude route. The result is not the assertion that off-line coordinates are illegal. The result is a typed equivalence between structural line readout and critical-normalized amplitude order.

6.1 Structural roles and explicit-formula roles

Definition 6.1 (Structural RH role packet)**.** *For a fixed nontrivial zerohood instance ρ, the structural RH packet consists of*

$$A_\zeta(\rho), \qquad C_\zeta(\rho), \qquad K_{\text{gov}}(\rho), \qquad B_\zeta(\rho), \qquad L_{\text{crit}}(\rho).$$

Here $A_\zeta(\rho)$ is analytic zerohood in the critical strip, $C_\zeta(\rho)$ is carrier preservation in the fixed zeta carrier, $K_{\text{gov}}(\rho)$ is kernel-governed same-scope report standing, $B_\zeta(\rho)$ is the carrier-preservation bridge, and $L_{\text{crit}}(\rho)$ is critical-line readout.

Definition 6.2 (Explicit-formula bridge object)**.** *$B_\zeta^{\text{EF}}(\rho)$ is the explicit-formula instantiation of the structural carrier-preservation bridge. It records a route from zeta zerohood standing to critical-normalized amplitude readout on the same carrier. It requires:*

- *(B1) same-zero preservation: the contribution is attached to the same ρ, not a symmetry partner, averaged cloud, finite height class, or model analogue;*
- *(B2) route preservation: the branch is read through* $\text{OfficialEFRoute}(R, \zeta, \psi)$*;*
- *(B3) contribution preservation: the explicit-formula term class associated with ρ is the carrier of the amplitude report;*
- *(B4) normalization preservation: the critical normalization by $x^{1/2}$ is fixed;*
- *(B5) bridge certification: any pairing, smoothing, truncation, or cancellation used to neutralize a non-neutral contribution is itself certified as endpoint-neutral on the same carrier;*
- *(B6) UEAP status preservation: the report does not exceed zerohood, contribution, amplitude, and bridge support;*
- *(B7) ATS trajectory preservation: the same anchor, tensor, and skin decomposition is preserved through the route.*

Audit 6.1 (Bridge object anti-circularity)**.** *$B_\zeta^{\text{EF}}(\rho)$ does not contain $L_{\text{crit}}(\rho)$ by definition. It is a route- and report-preservation bridge. A proof may instantiate it in bridge-complete form by showing endpoint-neutral amplitude order for the same zerohood contribution. Once endpoint-neutral order is obtained, the projection theorem yields line readout. The line readout is downstream of bridge completion, not built into the name of the bridge.*

6.2 Critical-normalized order invariant

Definition 6.3 (Critical-normalized amplitude order). *For $\rho = \beta + i\gamma$, define the critical-normalized amplitude order of the associated explicit-formula contribution by*

$$\operatorname{ord}_{1/2}(\rho) := \lim_{x\to\infty} \frac{\log|x^{\beta-1/2}|}{\log x},$$

whenever the contribution class is fixed by $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$. *The corresponding order discriminator is*

$$D_{\mathrm{ord}}(\rho) :\Longleftrightarrow \operatorname{ord}_{1/2}(\rho) \neq 0.$$

Theorem 6.1 (Normalized amplitude order invariant). *Under* $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$*, if $\rho = \beta + i\gamma$, then*

$$\operatorname{ord}_{1/2}(\rho) = \beta - \frac{1}{2}.$$

Consequently,

$$L_{\mathrm{crit}}(\rho) \Longleftrightarrow \operatorname{ord}_{1/2}(\rho) = 0, \qquad \neg L_{\mathrm{crit}}(\rho) \Longleftrightarrow D_{\mathrm{ord}}(\rho).$$

Proof. For $x > 1$, $|x^{\beta-1/2}| = x^{\beta-1/2}$. Therefore

$$\frac{\log|x^{\beta-1/2}|}{\log x} = \frac{(\beta - 1/2)\log x}{\log x} = \beta - \frac{1}{2}.$$

The limit is constant in x, so it equals $\beta - 1/2$. The two equivalences follow from $L_{\mathrm{crit}}(\rho)$ being $\beta = 1/2$. □

Audit 6.2 (Order invariant is not a hidden estimate). *The theorem does not estimate $\psi(x) - x$, does not assert dominance of a zero contribution, and does not deny cancellation in the full explicit formula. It identifies the order class of the contribution associated with a fixed zero under the fixed route. Pairing and cancellation remain bridge issues, not reasons to erase the order invariant at support level.*

6.3 Zerohood-to-contribution equivalence

Definition 6.4 (Same-zero explicit-formula contribution). $\mathrm{SameZero}_{\mathrm{EF}}(\rho)$ *means that the explicit-formula endpoint model contains the contribution class associated with the same nontrivial zeta zero ρ, with the same carrier, same route, and same critical normalization.*

Theorem 6.2 (Zerohood-to-contribution equivalence). *Under* $C_{\text{RH-EF}}(R, \zeta, \psi) \wedge \mathrm{OfficialEFRoute}(R, \zeta, \psi) \wedge \mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$,

$$A_\zeta(\rho) \Longleftrightarrow \mathrm{SameZero}_{\mathrm{EF}}(\rho)$$

for the endpoint-role purposes of this proof.

Proof. The forward direction is the zero-coverage clause of $\mathrm{Model}_{\mathrm{EF}}$: every nontrivial zerohood instance in the official carrier has an associated explicit-formula contribution class. The reverse direction is contribution soundness and same-zero identity: every contribution used by the endpoint model is attached to the same zeta zerohood carrier and the same zero instance. The equivalence is role-equivalence for the endpoint route; it does not add an analytic zero-location theorem. □

Audit 6.3 (No surrogate contribution). *A symmetry partner, random-matrix eigenvalue, spectral model point, finite verification entry, or smoothed auxiliary residue may support a bridge, but it is not* $\mathrm{SameZero}_{\mathrm{EF}}(\rho)$ *unless same-zero identity and carrier preservation are certified. This is the ATS anchor condition applied to the EF route.*

6.4 Line-readout-to-neutral-order bridge

Theorem 6.3 (Line readout equals neutral order on the EF route)**.** *Under* $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$*, the structural critical-line readout role and the EF neutral-order role are equivalent:*

$$L_{\mathrm{crit}}(\rho) \iff \big(\mathrm{ord}_{1/2}(\rho) = 0\big) \iff \mathrm{CritAmpOcc}(\rho).$$

The off-line branch is likewise equivalent to non-neutral order:

$$\neg L_{\mathrm{crit}}(\rho) \iff D_{\mathrm{ord}}(\rho) \iff \mathrm{OffLineAmpExcl}(\rho).$$

Proof. The first equivalence is the normalized order theorem. The second equivalence is the definition of critical-neutral amplitude occupation. The off-line equivalence is the exact complement: non-critical readout is nonzero normalized order, which is precisely off-line amplitude exclusion in the fixed EF model. □

Audit 6.4 (Equivalence is projection, not decision)**.** *The theorem fixes the role-content of the branches under* OfficialEFRoute*. It does not decide whether* $\mathrm{ord}_{1/2}(\rho)$ *is zero for every zero. That decision is obtained only after the local endpoint-countercase route is closed.*

6.5 Bridge completeness and bridge failure

Definition 6.5 (EF bridge completion)**.** $\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho)$ *means that the EF route supplies a same-zero, carrier-preserving, UEAP-adequate, ATS-stable bridge that neutralizes the endpoint-facing order contribution for* ρ *in the critical-normalized amplitude image. Formally, it entails*

$$\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho) \Rightarrow \mathrm{ord}_{1/2}(\rho) = 0.$$

$\mathrm{BridgeFail}_{\mathrm{EF}}(\rho)$ *means that no such endpoint-neutral bridge is supplied for the endpoint-facing use under audit.*

Theorem 6.4 (Bridge-complete EF route yields critical-line readout)**.** *Under* $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$*,*

$$\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho) \Rightarrow L_{\mathrm{crit}}(\rho).$$

Proof. Bridge completion yields endpoint-neutral normalized order by definition of the EF bridge-completion role. The line-readout-to-neutral-order theorem identifies neutral order with $L_{\mathrm{crit}}(\rho)$ on the fixed EF model. Hence critical-line readout follows. The proof does not define the bridge as line readout; it uses the EF order interface as the bridge output. □

Corollary 6.1 (Off-line assumption excludes bridge-complete neutralization)**.** *Under* $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$*,*

$$\neg L_{\mathrm{crit}}(\rho) \Rightarrow \neg\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho).$$

Proof. If $\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho)$, then the previous theorem gives $L_{\mathrm{crit}}(\rho)$, contradicting $\neg L_{\mathrm{crit}}(\rho)$. Therefore bridge-complete neutralization is unavailable inside the exact-complement off-line subproof. □

Audit 6.5 (Why bridge failure is not assumed globally)**.** *The corollary is local to the off-line reductio branch. The manuscript does not globally assert that EF bridge completion fails. The final proof instead shows that the off-line branch cannot survive: if a bridge completes, line readout follows; if it does not and endpoint counterforce remains on the fixed carrier, the non-neutral order becomes discriminator work.*

6.6 Bridge-discriminator-domain fork

Theorem 6.5 (Structural-to-EF bridge fork)**.** *Assume* $A_\zeta(\rho)$, $\mathrm{OfficialEFRoute}(R, \zeta, \psi)$, $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$, *and* $\mathrm{ProjectionRoleContent}_{\mathrm{EF}}(\neg L_{\mathrm{crit}}(\rho), \rho)$*. If the projected off-line branch is used with endpoint counterforce, then exactly one of the following endpoint-route dispositions applies:*

$$\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho) \quad \vee \quad D_{\mathrm{amp}}(\rho) \quad \vee \quad \mathrm{DomainShift}_{\mathrm{EF}}(\rho).$$

Proof. Projection-role content binds the local negative branch to non-neutral order content on the EF amplitude carrier. Endpoint counterforce says that this content is used to defeat critical-neutral endpoint closure. There are three possible dispositions. First, a bridge may neutralize the order content in the same endpoint image; this is $\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho)$. Second, the non-neutral order may remain endpoint-standing on the same fixed image. Then it is not support, bookkeeping, or skin; it changes branch exclusion and endpoint standing. By the endpoint-discriminator trichotomy, it is independent same-domain amplitude discriminator work, i.e. $D_{\mathrm{amp}}(\rho)$. Third, the act may avoid same-domain discrimination only by changing the endpoint image, carrier, route, normalization, or same-zero identity; that is $\mathrm{DomainShift}_{\mathrm{EF}}(\rho)$. UEAP excludes overreporting a support-level non-neutral order as endpoint result, and ATS excludes hiding a carrier or tensor change as notation skin. These cases exhaust the route. □

Audit 6.6 (The fork is the equivalence bridge)**.** *This is the bridge supplied by the structural-equivalence layer. It does not say non-neutral order is impossible. It says endpoint-facing non-neutral order has only three lawful dispositions: bridge-complete neutralization, domain shift, or independent discriminator work. The local off-line reductio eliminates bridge completion; fixed-domain endpoint use eliminates domain shift; the remaining branch is the amplitude discriminator.*

Corollary 6.2 (Fixed-domain off-line counterforce yields the amplitude discriminator)**.** *Under* $A_\zeta(\rho)$, $\mathrm{OfficialEFRoute}(R, \zeta, \psi)$, $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$, $\mathrm{FixedDomain}(R)$, $\mathrm{ProjectionRoleContent}_{\mathrm{EF}}(\neg L_{\mathrm{crit}}(\rho), \rho)$, $\mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho)$*, and the local off-line assumption* $\neg L_{\mathrm{crit}}(\rho)$*, one has*

$$D_{\mathrm{amp}}(\rho).$$

Proof. By the structural-to-EF bridge fork, the endpoint-counterforce use yields $\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho)$, $D_{\mathrm{amp}}(\rho)$, or $\mathrm{DomainShift}_{\mathrm{EF}}(\rho)$. The local off-line assumption excludes $\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho)$, since bridge completion yields $L_{\mathrm{crit}}(\rho)$. Fixed-domain preservation excludes $\mathrm{DomainShift}_{\mathrm{EF}}(\rho)$. Therefore $D_{\mathrm{amp}}(\rho)$ remains. □

Audit 6.7 (Hostile-referee target after the bridge)**.** *A successful objection must now attack a specific branch of the fork: show a bridge-complete neutralization compatible with* $\neg L_{\mathrm{crit}}(\rho)$*, show a same-domain endpoint use that is a domain shift and yet not a domain shift, or exhibit a fourth endpoint-standing role for non-neutral order that is not support, bookkeeping, bridge completion, domain shift, or independent discriminator work.*

7 Native Counterforce Collapse: No Fifth Endpoint Role

This section closes the remaining hostile-referee target left after the structural-to-explicit-formula bridge. A critic may grant that the off-line branch projects to non-neutral explicit-formula order, but still insist that this is simply native same-carrier falsification rather than illicit endpoint discrimination. The present section proves that ordinary native falsification is not a fifth endpoint role. Native analytic origin is preserved; endpoint-standing authority is classified.

7.1 Native analytic content versus native endpoint authority

Definition 7.1 (Native analytic content). *A datum is native analytic content for the RH-EF route when it is produced inside the fixed zeta/prime-trace carrier by the official explicit-formula model. In particular, if $\rho = \beta + i\gamma$ is a nontrivial zero, then the order class*

$$\mathrm{ord}_{1/2}(\rho) = \beta - \frac{1}{2}$$

is native analytic content of the associated contribution class.

Definition 7.2 (Native endpoint counterforce). $\mathrm{NativeCounterforce}_{\text{RH-EF}}(\rho)$ *means that native explicit-formula non-neutral order associated with the same zerohood instance ρ is used with endpoint-standing force to defeat the critical-neutral RH-EF endpoint while preserving the same official route, same zero, same prime-trace carrier, and same act-time endpoint evaluation. In this manuscript, that endpoint-standing force is supplied only by* $\mathrm{OfficialCounterexampleForce}_R(\mathrm{RH}^-_{\mathrm{bare}})$, $\mathrm{RHOffLineCounterexampleForce}(\rho)$, *or the local exact-complement force* $\mathrm{LocalCounterexampleForce}_{\text{RH-EF}}(\neg L_{\mathrm{crit}}(\rho); L_{\mathrm{crit}}(\rho))$ *together with* $\mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho)$. *Native counterforce is therefore not a raw property of* $\mathrm{ord}_{1/2}(\rho)$*; it is a use role.*

Audit 7.1 (The point of the definition). *The definition does not deny that* $\mathrm{ord}_{1/2}(\rho) \neq 0$ *is native analytic content. It isolates the stronger use: the native datum is no longer only descriptive or support-level; it is being used as endpoint-standing counterforce against the critical-neutral endpoint image. The proof below classifies that use. It does not outlaw the datum.*

Lemma 7.1 (Native content is not native authority). *Native analytic origin does not by itself supply endpoint-standing authority. In symbols,*

$$D_{\mathrm{ord}}(\rho) \not\Rightarrow \mathrm{NativeCounterforce}_{\text{RH-EF}}(\rho).$$

Proof. $D_{\mathrm{ord}}(\rho)$ states that the associated contribution has non-neutral critical-normalized order. It does not state that the datum resolves the endpoint, defeats the endpoint, excludes the positive branch, licenses downstream reuse, or stands as official endpoint result. UEAP separates support-level report from endpoint-standing report, and ATS separates tensor content from the act that uses that tensor content as endpoint authority. Therefore native analytic content alone is not native endpoint authority. □

7.2 Ordinary native off-line falsification package

This subsection isolates the strongest hostile reading of the off-line branch. The objector may say: the non-neutral order is not a separator or a second classifier; it is simply ordinary native falsification of RH. The answer is role-sensitive. Native falsification below endpoint use is preserved. Deployed native falsification, if it defeats the official endpoint on the same carrier, enters endpoint-use discipline and the endpoint-status trichotomy.

Definition 7.3 (Native off-line falsification package). *A native off-line falsification package $O_{\mathrm{nat}}(\rho)$ is a proposed endpoint role carrying the same nontrivial zerohood instance $A_\zeta(\rho)$, the same point $\rho = \beta + i\gamma$, the same zeta carrier, the same official explicit-formula route, the same contribution class attached to ρ, the same critical normalization by $x^{1/2}$, the same critical-normalized order* $\mathrm{ord}_{1/2}(\rho) = \beta - 1/2$, *the negative content $\neg L_{\mathrm{crit}}(\rho)$ equivalently* $\mathrm{ord}_{1/2}(\rho) \neq 0$, *and claimed endpoint-defeating force against the official RH endpoint.*

Audit 7.2 (Purpose of $O_{\text{nat}}(\rho)$)**.** *The package is not automatically illicit. It is a named worksheet for the hostile objection that off-line non-neutral order is ordinary native falsification. The proofs below classify the use of the package; they do not outlaw the coordinate fact or the explicit-formula contribution.*

Lemma 7.2 (Native off-line falsification below endpoint use)**.** *If $O_{\text{nat}}(\rho)$ is not deployed to defeat the official RH endpoint, it remains below endpoint use: raw content, ordinary theoremhood, coordinate description, support-level explicit-formula data, bookkeeping, or local hypothesis.*

Proof. Without endpoint-defeating deployment, the package does not exclude the positive endpoint branch, license downstream reuse as negative endpoint material, settle the official target, or change endpoint standing. It may remain mathematically meaningful as a coordinate report, explicit-formula support datum, local assumption, obstruction, or ordinary theorem. Those roles are lawful, but they do not yet perform endpoint resolution. □

Lemma 7.3 (Deployment fixes official RH counterexample force)**.** *If $O_{\text{nat}}(\rho)$ is deployed to defeat the official RH endpoint on the fixed RH-EF carrier, then* $\mathrm{RHOffLineCounterexampleForce}(\rho)$ *holds.*

Proof. Deployment to defeat the endpoint means that the package is not merely recorded as coordinate or amplitude support. It is used against the official positive branch. Because the package preserves the same zero, zeta carrier, explicit-formula route, contribution class, critical normalization, and non-neutral order, the force is same-carrier off-line counterexample force for the RH-EF endpoint. That is exactly $\mathrm{RHOffLineCounterexampleForce}(\rho)$. □

Lemma 7.4 (Deployed native off-line falsification is endpoint use)**.** *A deployed native off-line falsification package is endpoint use:*

$$O_{\text{nat}}(\rho) \textit{ deployed as endpoint defeat} \Rightarrow \mathrm{EndpointUse}_R(\neg L_{\text{crit}}(\rho)).$$

Proof. By the previous lemma, deployed $O_{\text{nat}}(\rho)$ is official RH off-line counterexample force. Official counterexample force is endpoint use by the general counterexample-force theorem and by the RH-EF specialization. Therefore the deployed package is endpoint use of the off-line branch. □

Lemma 7.5 (Endpoint-use native off-line falsification enters the endpoint-status trichotomy)**.** *A deployed native off-line falsification package that is endpoint use is bookkeeping, governance-equivalent to the positive critical-neutral amplitude interface, independent same-domain amplitude endpoint-status work, or domain/interface shift.*

Proof. Endpoint use fixes the carrier, route, normalization, branch role, and endpoint image. A status factor attached to that use either changes no endpoint-governance verdict, in which case it is bookkeeping; reproduces the admitted positive interface, in which case it is governance-equivalent to neutral occupation; changes endpoint status on the same carrier without being equivalent, in which case it is independent same-domain amplitude status work; or changes the admissibility-relevant carrier, route, normalization, or endpoint image, in which case it is domain/interface shift. These cases exhaust whether the alleged role changes endpoint work and whether it remains on the fixed carrier. □

Lemma 7.6 (Deployed non-neutral amplitude is not governance-equivalent to neutral occupation)**.** *A deployed native off-line falsification package cannot be governance-equivalent to critical-neutral amplitude occupation while preserving endpoint-defeating negative force.*

Proof. The positive endpoint interface is $\mathrm{ord}_{1/2}(\rho) = 0$, equivalently critical-neutral amplitude occupation. The deployed off-line package asserts and uses $\mathrm{ord}_{1/2}(\rho) \neq 0$ to defeat that interface. If it were governance-equivalent to neutral occupation, it would not exclude the positive endpoint branch. If it excludes the positive endpoint branch, it performs a different endpoint-status role. Therefore, while retaining endpoint-defeating force, deployed non-neutral amplitude is not governance-equivalent to the positive interface. □

Theorem 7.1 (Ordinary native off-line falsification collapse, expanded form)**.** *A native off-line falsification package $O_{\mathrm{nat}}(\rho)$ has exactly the following same-mode possibilities: it is raw, support, local, bookkeeping, or theorem-level content below endpoint use; it is bridge-neutralized; it is bookkeeping; it is a domain/interface shift; or it is the independent amplitude discriminator $D_{\mathrm{amp}}(\rho)$. There is no autonomous ordinary-negative endpoint role outside this list.*

Proof. If the package is not deployed to defeat the endpoint, it remains below endpoint use by the first lemma. If a same-zero, same-route bridge neutralizes the non-neutral order in the critical-normalized image, it is bridge-neutralized. If it changes no endpoint-relevant feature, it is bookkeeping. If it changes the zero, zeta carrier, contribution class, route, normalization, prime-trace carrier, or endpoint image, it is a domain/interface shift. It remains to consider a package deployed to defeat the endpoint while preserving the fixed carrier and while not being bridge-neutralized or bookkeeping. By the deployment lemmas, it is endpoint use. By the trichotomy lemma, it is bookkeeping, governance-equivalent to neutral occupation, independent same-domain amplitude work, or domain/interface shift. Bookkeeping and domain/interface shift have already been removed. Governance-equivalence to neutral occupation is impossible while the package preserves endpoint-defeating non-neutral force. The remaining case is independent same-domain amplitude endpoint-status work, namely $D_{\mathrm{amp}}(\rho)$. □

Corollary 7.1 (No sixth role in the live RH-EF subproof)**.** *Inside the exact-complement reductio subproof, the opened assumption $[\neg L_{\mathrm{crit}}(\rho)]_i$ cannot be deployed as ordinary native endpoint falsification without yielding $D_{\mathrm{amp}}(\rho)$.*

Proof. The opened assumption is the live exact-complement countercase against $L_{\mathrm{crit}}(\rho)$. Once it is used with endpoint counterforce on the fixed RH-EF carrier, it is deployed $O_{\mathrm{nat}}(\rho)$, not inert syntax. Fixed-domain evaluation removes domain/interface shift. Endpoint counterforce removes support-only and bookkeeping readings. The live off-line assumption removes bridge-neutralized positive occupation. By the expanded collapse theorem, the remaining endpoint role is $D_{\mathrm{amp}}(\rho)$. □

Audit 7.3 (Technical closure achieved)**.** *After this subsection, the objection "an off-line zero is just ordinary native falsification" is not an unaddressed case. It is a challenge to the deployed-native-falsification theorem chain: deployment fixes official counterexample force; official counterexample force is endpoint use; endpoint-use native falsification enters the trichotomy; and deployed non-neutral amplitude is not governance-equivalent to neutral endpoint occupation.*

7.3 The native-counterforce fork

Definition 7.4 (Bridge-neutralized native counterforce)**.** $\mathrm{BridgeNeutralized}(\rho)$ *means that the apparent non-neutral endpoint-facing contribution has a same-zero, same-route, carrier-preserving bridge that neutralizes its endpoint order in the critical-normalized image. Bridge-neutralization is stronger than cancellation as a numerical or formal possibility: it is endpoint-neutralization certified at the same standing-bearing role level.*

Theorem 7.2 (Native-counterforce role exhaustion)**.** *For the fixed RH-EF endpoint carrier, native endpoint counterforce by non-neutral order has no fifth role. If*

$$A_\zeta(\rho) \wedge \mathrm{OfficialEFRoute}(R, \zeta, \psi) \wedge \mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R) \wedge \mathrm{NativeCounterforce}_{\text{RH-EF}}(\rho),$$

then exactly one of the following endpoint-role dispositions applies:

$$\mathrm{BridgeNeutralized}(\rho) \vee \mathrm{SupportOnly}(\rho) \vee \mathrm{Bookkeeping}(\rho) \vee \mathrm{DomainShift}_{\mathrm{EF}}(\rho) \vee D_{\mathrm{amp}}(\rho).$$

Proof. The datum under audit is native analytic content because it is produced by the fixed explicit-formula route. The question is what role it plays when used with endpoint-standing counterforce.

If a same-zero, same-route bridge proves endpoint-neutralization of the non-neutral contribution at the critical-normalized endpoint image, then BridgeNeutralized(ρ) holds. If the datum is used only as part of an argument, obstruction, heuristic, or local analytic description, then it is SupportOnly(ρ), and it lacks endpoint-standing counterforce. If the datum changes no endpoint standing, reportability, downstream reuse, branch exclusion, or admissible continuation, then it is bookkeeping. If the datum obtains endpoint force only by changing the zeta carrier, the prime-trace route, the normalization, same-zero identity, the smoothing/truncation regime as tensor authority, or the endpoint image, then $\mathrm{DomainShift}_{\mathrm{EF}}(\rho)$ holds.

After these cases are removed, the remaining use is endpoint-standing, same-domain, non-neutral, non-support, non-bookkeeping, non-bridge-neutralized, and non-domain-shifting. It changes whether the live off-line branch can defeat critical-neutral endpoint closure. By the endpoint-discriminator trichotomy and the ATS no-second-classifier lock, that remaining use is independent same-domain amplitude endpoint-status discrimination. This is exactly $D_{\mathrm{amp}}(\rho)$. The cases are exhaustive because every endpoint-facing use either changes endpoint governance or it does not; if it changes governance, it either preserves the invariant bundle or changes it. □

Audit 7.4 (Ordinary native falsification is not a fifth role)**.** *The theorem answers the strongest hostile objection. An off-line zero's non-neutral order is native analytic content. But when that content is used as endpoint-standing counterforce, ordinary native falsification is not a role outside the endpoint taxonomy. It is bridge-neutralized, support-only, bookkeeping, domain/interface shift, or independent endpoint-status discrimination. The proof does not deny counterexample syntax; it classifies endpoint-standing counterexample force.*

7.4 Collapse under the live off-line countercase

Lemma 7.7 (Support-only native content has no endpoint counterforce)**.**

$$\mathrm{SupportOnly}(\rho) \Rightarrow \neg\mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho).$$

Proof. Support-only material may contribute to a proof, bridge program, heuristic, obstruction, or local analytic diagnosis, but it does not itself defeat the official endpoint. Endpoint counterforce is precisely the role in which a branch blocks endpoint closure. Therefore support-only use and endpoint counterforce are incompatible. □

Lemma 7.8 (Bookkeeping native content has no endpoint counterforce)**.**

$$\mathrm{Bookkeeping}(\rho) \Rightarrow \neg\mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho).$$

Proof. Bookkeeping changes no endpoint standing, reportability, downstream reuse, branch exclusion, or admissible continuation. Endpoint counterforce changes exactly the endpoint-facing status of the live branch. Hence bookkeeping cannot carry endpoint counterforce. □

Lemma 7.9 (Bridge-neutralization is incompatible with the live off-line endpoint countercase)**.** *Under* $\mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R)$*, the exact-complement off-line subproof satisfies*

$$\neg L_{\mathrm{crit}}(\rho) \Rightarrow \neg \mathrm{BridgeNeutralized}(\rho).$$

Proof. Bridge-neutralization is endpoint-level neutralization of the same-zero non-neutral order in the critical-normalized image. In the fixed EF model, neutral endpoint order is equivalent to $L_{\mathrm{crit}}(\rho)$. Thus $\mathrm{BridgeNeutralized}(\rho)$ gives the same endpoint effect as bridge-complete neutralization: $L_{\mathrm{crit}}(\rho)$. This contradicts the live off-line assumption $\neg L_{\mathrm{crit}}(\rho)$. The argument is local to the reductio subproof; it does not globally deny that bridge-neutralization may be proved for a zero. □

Theorem 7.3 (Native same-carrier counterforce collapse)**.** *Under the fixed RH-EF endpoint context,*

$$\begin{aligned} &A_\zeta(\rho) \wedge \mathrm{OfficialEFRoute}(R, \zeta, \psi) \wedge \mathrm{Model}_{\mathrm{EF}}(\zeta, \psi, R) \wedge \mathrm{FixedDomain}(R) \\ &\quad \wedge \mathrm{ProjectionRoleContent}_{\mathrm{EF}}(\neg L_{\mathrm{crit}}(\rho), \rho) \wedge \mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho) \wedge \neg L_{\mathrm{crit}}(\rho) \Rightarrow D_{\mathrm{amp}}(\rho). \end{aligned}$$

Proof. Projection-role content and endpoint counterforce make the non-neutral EF order into native endpoint counterforce, so $\mathrm{NativeCounterforce}_{\text{RH-EF}}(\rho)$ holds. By native-counterforce role exhaustion, the role is bridge-neutralized, support-only, bookkeeping, domain shift, or $D_{\mathrm{amp}}(\rho)$. Bridge-neutralization is excluded by the live off-line assumption. Support-only and bookkeeping are incompatible with endpoint counterforce. Domain shift is excluded by $\mathrm{FixedDomain}(R)$. The remaining exhaustive branch is $D_{\mathrm{amp}}(\rho)$. □

Audit 7.5 (Final hardening of the hinge)**.** *The theorem is the sharpened hinge. It does not say* $\neg L_{\mathrm{crit}}(\rho) \Rightarrow D_{\mathrm{amp}}(\rho)$*. It says that once the off-line branch is projected through the official EF route and used with endpoint counterforce on the fixed carrier, the ordinary-native-falsification reading collapses into the same endpoint taxonomy: bridge-neutralized, support-only, bookkeeping, domain shift, or independent discriminator. The first four branches are incompatible with the live local countercase and fixed-domain endpoint use, so the discriminator remains.*

Corollary 7.2 (No native-falsification escape from A+ closure)**.** *There is no same-carrier endpoint-standing role in which non-neutral EF order defeats RH while remaining neither bridge-neutralized, support-only, bookkeeping, domain-shifting, nor an independent amplitude discriminator.*

Proof. This is the contrapositive of native-counterforce role exhaustion together with the collapse theorem. A purported native-falsification escape that changes endpoint standing must occupy one of the enumerated roles. If it avoids all enumerated roles, it changes no endpoint standing and is endpoint-inert. □

8 Prime-Trace Neutrality and Non-Neutral Scale Standing

8.1 Descriptive amplitude versus endpoint-standing amplitude

Definition 8.1 (Amplitude status levels)**.** *For a zerohood instance* ρ*, distinguish:*

1. descriptive amplitude*: the algebraic scale class* $x^{\beta-1/2}$*;*

2. support-level amplitude*: use of the scale class in an analytic argument or bridge program;*
3. endpoint-standing amplitude*: use of the scale class to defeat or settle the official RH endpoint on the fixed explicit-formula carrier.*

Definition 8.2 (Endpoint-standing non-neutral amplitude)**.** EndpointStandingNonNeutralAmp(ρ) *means that the non-neutral scale $x^{\beta-1/2} \neq x^0$ is used as endpoint-standing content against the critical-neutral amplitude image while preserving the same zeta/prime-trace carrier.*

Theorem 8.1 (Descriptive non-neutral amplitude is lawful)**.**

$$\mathrm{OffLineAmpExcl}(\rho) \not\Rightarrow \mathrm{EndpointStandingNonNeutralAmp}(\rho).$$

Descriptive or support-level non-neutral amplitude remains lawful analytic content.

Proof. The existence of a scale class is not the same as endpoint-standing use. UEAP separates the support report from the endpoint report. ATS separates tensor content from the act that uses that tensor content as endpoint authority. Therefore non-neutral amplitude by itself does not become endpoint-standing amplitude. □

8.2 Prime-trace endpoint adequacy

Definition 8.3 (Prime-trace endpoint adequacy)**.** PrimeTraceEndpointAdeq(R, ζ, ψ) *means that the explicit-formula route and critical-normalized prime-trace image are adequate for the official RH endpoint: critical-line readout projects to neutral amplitude occupation, off-line readout projects to non-neutral amplitude non-occupation, and the projection preserves the same zeta zerohood carrier. It does not assert universal neutral occupation.*

Theorem 8.2 (Neutral amplitude is endpoint image, not assumption)**.** PrimeTraceEndpointAdeq(R, ζ, ψ) *fixes neutral amplitude as the positive endpoint image, but does not assert that every zero contribution occupies that image. In symbols,*

$$\mathrm{PrimeTraceEndpointAdeq}(R, \zeta, \psi) \not\Rightarrow \forall\rho(A_\zeta(\rho) \Rightarrow \mathrm{CritAmpOcc}(\rho)).$$

Proof. Endpoint adequacy specifies what occupation would mean if the positive branch is occupied. It is a typing condition, not an existence or universal occupation condition. The distinction is the same as fixing a candidate endpoint image without asserting a non-failing candidate in the SAT setting. □

8.3 Non-neutral scale standing

Theorem 8.3 (Non-neutral scale standing theorem)**.** *If non-neutral amplitude is used with endpoint-standing force on the fixed RH-EF carrier, then it performs same-domain noncritical amplitude status work:*

$$\begin{aligned}&\mathrm{EndpointStandingNonNeutralAmp}(\rho) \wedge \mathrm{FixedDomain}(R)\\ &\quad \wedge \mathrm{OfficialEFRoute}(R, \zeta, \psi) \Rightarrow \mathrm{NonCriticalAmpStatusWork}_R(\rho).\end{aligned}$$

Proof. Endpoint-standing non-neutral amplitude is not merely descriptive because it is used to decide endpoint standing. It is not support-only because it defeats or displaces the endpoint. It is not bookkeeping because removing it changes endpoint counterforce. It is not carrier shift because the fixed-domain premise preserves the zeta/prime-trace carrier. It is not repair because the standing is asserted at the local act time. It is not coequal occupation because the single amplitude interface says non-neutral scale is non-occupation of the critical-neutral image. Therefore it is same-domain noncritical amplitude status work. □

Audit 8.1 (No RH-equivalent bound smuggled). *The theorem does not assert a bound such as* $\psi(x)-x = O(x^{1/2+\epsilon})$*, nor does it deny that a hypothetical off-line zero would affect the prime trace. It classifies only the endpoint-standing use of the non-neutral scale. Descriptive and support uses remain lawful.*

Theorem 8.4 (Non-native amplitude theorem). *On the fixed critical-neutral amplitude image, same-domain noncritical amplitude status work is non-native as endpoint authority unless it is support, bookkeeping, carrier shift, repair, surrogate substitution, or lawful coequal occupation. Consequently,*

$$\mathrm{NonCriticalAmpStatusWork}_R(\rho) \Rightarrow \mathrm{NonNativeAmpDisc}(\rho).$$

Proof. The adjective "native" is role-relative. The scale exponent is native analytic data in the explicit formula. It is not native as endpoint authority over the critical-neutral image when it claims to defeat that image while preserving the same carrier. If the non-neutral scale is only descriptive, it is not endpoint-status work. If it is support, it does not resolve the endpoint. If it changes the carrier or route, it is not same-domain. If it is repair, it violates act-time finality. If it is coequal occupation, it has changed to the broad outcome carrier. After these lawful roles are removed, the remaining role is an independent amplitude endpoint-status discriminator. □

Audit 8.2 (Native-data objection). *A classical analyst may correctly say that* $x^{\beta-1/2}$ *is native analytic data. The manuscript agrees. The excluded object is not the data but its endpoint-standing use as same-carrier authority over the critical-neutral image. This distinction is what prevents the proof from banning analytic content.*

9 Amplitude Separator Occupation and Discriminator Closure

9.1 Separator occupation

Definition 9.1 (Local amplitude separator occupation). $\mathrm{AmpSep}^{\mathrm{loc}}(\rho)$ *means that off-line amplitude non-occupation is used inside the live exact-complement reductio subproof as the endpoint-defeating negative branch on the fixed RH-EF carrier. It is local separator occupation, not global endpoint outcome.*

Definition 9.2 (Global amplitude separator occupation). $\mathrm{AmpSep}^{\mathrm{occ}}(\rho)$ *means that off-line amplitude non-occupation is used as the official endpoint-resolving negative branch for RH on the fixed RH-EF carrier.*

Theorem 9.1 (Separator occupation is endpoint-standing non-neutral amplitude).

$$\mathrm{AmpSep}^{\mathrm{loc}}(\rho) \Rightarrow \mathrm{EndpointStandingNonNeutralAmp}(\rho),$$
$$\mathrm{AmpSep}^{\mathrm{occ}}(\rho) \Rightarrow \mathrm{EndpointStandingNonNeutralAmp}(\rho).$$

Proof. By definition, separator occupation is the endpoint-facing use of non-neutral amplitude non-occupation. Local separator occupation has this role inside the reductio subproof; global separator occupation has it as official negative endpoint resolution. Both are endpoint-standing uses rather than descriptive or support reports. □

9.2 Endpoint-discriminator trichotomy

Definition 9.3 (Amplitude endpoint-status discriminator). *C is an amplitude endpoint-status discriminator for ρ when it changes endpoint standing, branch exclusion, reportability, downstream reuse, or admissible continuation for the fixed RH-EF endpoint act involving ρ.*

Theorem 9.2 (Endpoint-discriminator trichotomy). *Every amplitude endpoint-status classifier attached to a fixed RH-EF endpoint act is exactly one of: bookkeeping, governance-equivalent to the admitted interface, independent same-domain endpoint-status work, or domain shift.*

Proof. If the classifier changes no endpoint verdict, it is bookkeeping. If it reproduces the existing admitted interface, it is governance-equivalent. If it changes endpoint-status verdicts while preserving the same invariant bundle, it is independent same-domain work. If it changes the invariant bundle, it is domain shift. The cases are exhaustive by whether governance work changes and whether the change remains on the same fixed domain. □

Theorem 9.3 (Hinge theorem: noncritical amplitude work induces the amplitude discriminator).

$$\mathrm{NonCriticalAmpStatusWork}_R(\rho) \Rightarrow D_{\mathrm{amp}}(\rho).$$

Proof. Noncritical amplitude work is endpoint-facing, same-carrier, non-neutral amplitude status determination. It is not positive critical-neutral occupation. It is not support-only, bookkeeping, repair, carrier shift, surrogate substitution, or coequal occupation by the hypotheses built into NonCriticalAmpStatusWork. By the endpoint-discriminator trichotomy, the only remaining role is independent same-domain endpoint-status work. In the explicit-formula amplitude carrier, that role is exactly $D_{\mathrm{amp}}(\rho)$. □

Audit 9.1 (This is the central hostile-referee target). *The theorem does not say that non-neutral amplitude is impossible. It says that noncritical amplitude* status work*, after support/bookkeeping/carrier-shift/repair/coequal roles are excluded, is independent endpoint-status discrimination. A successful objection must produce a remaining lawful role for endpoint-standing non-neutral amplitude that is not independent discrimination.*

Theorem 9.4 (Endpoint use triggers no-independent amplitude closure).

$$\mathrm{EndpointUse}_R(\mathrm{AmpSep}^{\mathrm{loc}}(\rho)) \wedge \mathrm{FixedDomain}(R) \Rightarrow N_{\mathrm{amp}}(\rho).$$

Proof. Local separator occupation is endpoint use inside a governed reductio subproof. Endpoint use on the fixed domain instantiates the kernel and the A+ consequence layer. The local A+ consequence includes no independent same-domain endpoint-status discriminator on the same RH-EF carrier. This is the predicate $N_{\mathrm{amp}}(\rho)$. □

Theorem 9.5 (No-independent amplitude closure excludes the discriminator).

$$N_{\mathrm{amp}}(\rho) \Rightarrow \neg D_{\mathrm{amp}}(\rho).$$

Proof. N_{amp} is the no-independent-discriminator closure specialized to the amplitude endpoint-status discriminator. Therefore it denies D_{amp}. □

Corollary 9.1 (Local amplitude separator occupation is impossible).

$$\mathrm{AmpSep}^{\mathrm{loc}}(\rho) \Rightarrow \perp.$$

Proof. Assume $\mathrm{AmpSep}^{\mathrm{loc}}(\rho)$. Separator occupation gives endpoint-standing non-neutral amplitude. By the non-neutral scale standing theorem and the hinge theorem, $D_{\mathrm{amp}}(\rho)$. Endpoint use of the local separator gives $N_{\mathrm{amp}}(\rho)$, hence $\neg D_{\mathrm{amp}}(\rho)$. Contradiction. □

10 Anti-Trivialization Discipline

Theorem 10.1 (Exact-complement assumptions are not automatically separators)**.** *For an arbitrary proposition P, a local reductio assumption $[\neg P]_i$ does not become separator occupation merely because it is the exact complement of P.*

Proof. Separator occupation requires a fixed endpoint-image carrier, an official endpoint route, projection-role content on that carrier, endpoint-counterforce use, and exclusion of support, bookkeeping, carrier shift, repair, surrogate substitution, and coequal occupation. A bare local assumption has none of those features by itself. Therefore arbitrary negation is not separator occupation. □

Corollary 10.1 (RH-specific separator conditions)**.** *For RH-EF, $[\neg L_{\mathrm{crit}}(\rho)]_i$ becomes local amplitude separator occupation only after the official explicit-formula route binds it to non-neutral amplitude projection and endpoint counterforce uses that projection as endpoint-defeating content.*

Audit 10.1 (No $\neg P \Rightarrow P$ schema)**.** *The proof cannot be instantiated for arbitrary P. It uses the explicit-formula amplitude projection of zeta zerohood. Without such a projection and a proof that endpoint-standing non-neutral amplitude is independent discriminator work, the A+ contradiction cannot be triggered.*

11 Local Reductio Engine and Universal Endpoint

11.1 Exact-complement annotation

Definition 11.1 (Exact-complement reductio annotation)**.**

$$\mathrm{ReductioCountercase}_{\text{RH-EF}}(\neg L_{\mathrm{crit}}(\rho); L_{\mathrm{crit}}(\rho))$$

is the proof-act annotation generated when the sole local assumption $[\neg L_{\mathrm{crit}}(\rho)]_i$ is opened as the live exact complement of $L_{\mathrm{crit}}(\rho)$ under $A_\zeta(\rho)$ in the official RH-EF endpoint-closure proof. It is not an additional object-level premise.

Theorem 11.1 (Annotation is not a premise)**.** *The annotation does not transform the discharged assumption into*

$$\neg L_{\mathrm{crit}}(\rho) \wedge \mathrm{ReductioCountercase}_{\text{RH-EF}}(\neg L_{\mathrm{crit}}(\rho); L_{\mathrm{crit}}(\rho)).$$

It records the local proof role of the sole assumption $[\neg L_{\mathrm{crit}}(\rho)]_i$.

Proof. A reductio subproof has an object-level assumption and a proof role for that assumption. When the assumption is the exact complement of the target in a fixed endpoint proof, the proof role is live countercase use. The role annotation is metaproof bookkeeping for the subproof; it is not a formula added to the object language. □

Theorem 11.2 (Local countercase to endpoint counterforce). *Under* $A_\zeta(\rho) \wedge \mathrm{OSEval}_{\text{RH-EF}}(R, \zeta, \psi)$, *the exact-complement local annotation yields endpoint counterforce:*

$$\mathrm{ReductioCountercase}_{\text{RH-EF}}(\neg L_{\text{crit}}(\rho); L_{\text{crit}}(\rho)) \Rightarrow \mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho).$$

Proof. The live exact-complement assumption is introduced precisely to block the target $L_{\text{crit}}(\rho)$ under the official endpoint split. That is local endpoint-facing force, not global theoremhood and not official negative endpoint resolution. The endpoint-closure evaluation supplies the fixed target and branch slots required for this local role. □

Theorem 11.3 (Local exact-complement force is local endpoint use). *Under* $A_\zeta(\rho) \wedge \mathrm{OSEval}_{\text{RH-EF}}(R, \zeta, \psi)$, *the exact-complement reductio role and endpoint counterforce yield local endpoint use:*

$$\mathrm{ReductioCountercase}_{\text{RH-EF}}(\neg L_{\text{crit}}(\rho); L_{\text{crit}}(\rho)) \wedge \mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho) \Rightarrow \mathrm{LocalEndpointUse}_R(\neg L_{\text{crit}}(\rho)).$$

Proof. The reductio annotation supplies $\mathrm{LocalCounterexampleForce}_{\text{RH-EF}}(\neg L_{\text{crit}}(\rho); L_{\text{crit}}(\rho))$: the off-line branch is the live exact-complement countercase to the local target. Endpoint counterforce says that this branch is being used to block endpoint closure. The general local counterexample-force theorem, Theorem 1.6, therefore gives local endpoint use. This is still local; it does not assert global official negative endpoint resolution. □

Corollary 11.1 (Local A+ closure for the exact-complement countercase). *Under fixed-domain RH-EF evaluation,*

$$\mathrm{LocalEndpointUse}_R(\neg L_{\text{crit}}(\rho)) \wedge \mathrm{FixedDomain}(R) \Rightarrow A^+_{R,\text{loc}}(\neg L_{\text{crit}}(\rho)) \Rightarrow N_{\text{amp}}(\rho).$$

Proof. Apply Theorem 1.7. In the RH-EF route, the local A+ consequence relevant to amplitude separator use is the no-independent-amplitude-discriminator closure $N_{\text{amp}}(\rho)$. □

Theorem 11.4 (Endpoint counterforce induces local amplitude separator occupation). *Under the fixed RH-EF context,*

$$A_\zeta(\rho) \wedge \mathrm{ProjectionRoleContent}_{\text{EF}}(\neg L_{\text{crit}}(\rho), \rho) \wedge \mathrm{EndpointCounterforce}_{\text{RH-EF}}(\rho) \Rightarrow \mathrm{AmpSep}^{\text{loc}}(\rho).$$

Proof. Projection-role content says that the local negative branch has non-neutral amplitude non-occupation as its endpoint-role content. Endpoint counterforce says that this projected non-occupation is being used to defeat the critical-neutral endpoint image inside the local subproof. It is therefore local amplitude separator occupation. This is not a promotion of raw negation; it is the endpoint-use classification of the projected amplitude role under counterforce. □

Theorem 11.5 (Local countercase contradiction). *Assume* $A_\zeta(\rho)$, $C_{\text{RH-EF}}(R, \zeta, \psi)$, $\mathrm{OfficialEFRoute}(R, \zeta, \psi)$, $\mathrm{Model}_{\text{EF}}(\zeta, \psi, R)$, $\mathrm{FixedDomain}(R)$, *and* $\mathrm{OSEval}_{\text{RH-EF}}(R, \zeta, \psi)$. *Then the local exact-complement assumption* $[\neg L_{\text{crit}}(\rho)]_i$ *leads to contradiction.*

Proof. Open $[\neg L_{\text{crit}}(\rho)]_i$ as the live exact complement of $L_{\text{crit}}(\rho)$. The annotation theorem supplies the reductio-countercase role but no additional object-level premise. The local countercase has endpoint counterforce. By the local off-line projection theorem, it has projection-role content $\mathrm{ProjectionRoleContent}_{\text{EF}}(\neg L_{\text{crit}}(\rho), \rho)$. Endpoint counterforce plus projection-role content gives $\mathrm{AmpSep}^{\text{loc}}(\rho)$. The same local exact-complement use is local endpoint use, so local A+ closure supplies $N_{\text{amp}}(\rho)$. Local amplitude separator occupation is impossible by the discriminator contradiction. Hence the subproof derives $\perp$. □

Theorem 11.6 (Annotation discharge). *If the annotated exact-complement subproof derives contradiction from the sole object-level assumption* $[\neg L_{\text{crit}}(\rho)]_i$*, then* $L_{\text{crit}}(\rho)$ *follows. The discharged formula is* $\neg L_{\text{crit}}(\rho)$*, not a strengthened conjunction.*

Proof. The reductio annotation is a proof-act role, not an object-level assumption. Therefore ordinary reductio discharges the sole opened formula. Since the target is $L_{\text{crit}}(\rho)$, contradiction under $\neg L_{\text{crit}}(\rho)$ gives $L_{\text{crit}}(\rho)$. □

Theorem 11.7 (Arbitrary zerohood gives critical-line readout). *Under the fixed official RH-EF endpoint context,*

$$A_\zeta(\rho) \Rightarrow L_{\text{crit}}(\rho).$$

Proof. Assume $A_\zeta(\rho)$. If $L_{\text{crit}}(\rho)$, done. If $\neg L_{\text{crit}}(\rho)$, open the exact-complement local subproof. The local countercase contradiction discharges the assumption and yields $L_{\text{crit}}(\rho)$. Thus $A_\zeta(\rho) \Rightarrow L_{\text{crit}}(\rho)$. □

Corollary 11.2 (Riemann Hypothesis).

$$\forall\rho\,\big(A_\zeta(\rho) \Rightarrow L_{\text{crit}}(\rho)\big).$$

Proof. The preceding theorem holds for arbitrary ρ. Universal generalization yields the displayed formula, which is the Riemann Hypothesis in the standard critical-strip formulation. □

12 Ordinary Negative Theoremhood, Incompleteness, and Open Status

Theorem 12.1 (Ordinary negative theoremhood is not banned). *A hypothetical ordinary theorem of* $\mathrm{RH}^-_{\text{bare}}$ *is not rejected merely because it is negative. Ordinary theoremhood is proof legitimacy. It becomes endpoint-role disciplined only when used as official endpoint resolution or endpoint counterforce on the fixed carrier.*

Proof. The kernel applies to genuine theoremhood regardless of polarity. The excluded object is not negative theoremhood as such, but endpoint-standing non-neutral amplitude separator work. If a negative theorem is not used as official endpoint resolution, it is not the endpoint act under audit. If it is so used, ordinary official resolution supplies endpoint use and triggers the endpoint-role discipline. □

Theorem 12.2 (Official negative endpoint resolution becomes endpoint use).

$$\mathrm{OrdOfficialRes}_R(\mathrm{RH}^-_{\text{bare}}) \wedge \mathrm{OfficialEFRoute}(R,\zeta,\psi) \Rightarrow \mathrm{EndpointUse}_R(\mathrm{RH}^-_{\text{bare}}).$$

Proof. Ordinary official resolution fixes target, carrier, branch, theorem-status, downstream reuse, act-time finality, and redescription-stable endpoint role. Those are exactly endpoint-use conditions. The official explicit-formula route fixes how the negative branch is projected into amplitude content. □

Definition 12.1 (Same-domain incompleteness or independence objection). $\mathrm{SameDomainIncomp}_{\mathrm{RH}}(C)$ *means that a Godel-style, independence, true-but-unprovable, model-theoretic, stronger-theory, or proof-system distinction is asserted to re-enter the fixed RH-EF endpoint carrier as a standing-bearing distinction.*

Theorem 12.3 (Incompleteness firewall). *If a same-domain incompleteness or independence distinction does not affect endpoint standing, reportability, downstream reuse, branch exclusion, or admissible continuation, it is endpoint-inert. If it does affect one of those features, it is endpoint-admissibility-relevant gate work and is subject to the same A+ no-independent-discriminator closure.*

Proof. Endpoint force means altering the standing-bearing endpoint act. A distinction that does not alter the act is inert with respect to the endpoint. A distinction that does alter the act is a classifier of endpoint standing, reportability, reuse, or continuation. It therefore belongs to the endpoint-admissibility-relevant invariant bundle controlled by the kernel and A+ consequence layer. □

Corollary 12.1 (Open status is not a third endpoint branch). *Open, unresolved, independent, unproved, or review status is epistemic status. It is not a third endpoint truth branch of the RH-EF endpoint.*

13 Route-Locus Exhaustion and Weakening Resistance

13.1 Weakening-resistance of the local RH-EF kernel packet

The local K-packet prevents a hidden weaker same-carrier regime from admitting endpoint-standing non-neutral amplitude while preserving the appearance of the same RH-EF endpoint. The result is local: plural analytic techniques are preserved as support or bridge material, but they cannot become a second endpoint interior for the same critical-neutral amplitude slot.

Definition 13.1 (Core RH-EF endpoint adequacy packet). *For a fixed zerohood instance ρ, let*

$$C^0_{\text{RH-EF}}(\rho) = \{Ref_\rho, St_\rho, Adm_\rho, Irr_\rho\}$$

together with target determinacy, step evaluability, act-time finality, same-regime fidelity, the fixed zeta carrier, the fixed nontrivial zerohood instance $A_\zeta(\rho)$, same-zero identity, the fixed explicit-formula route, the fixed prime-trace carrier, the fixed critical normalization, and the fixed critical-neutral amplitude endpoint image. This is the core packet already forced before the full local K1–K13 consequence layer is invoked.

Definition 13.2 (Non-neutral-amplitude-governance-permissive weakening). *A non-neutral-amplitude-governance-permissive weakening $W_{\text{RH-EF}}(W, \rho)$ is a regime W satisfying the following conditions:*

1. *it preserves the same zero ρ;*
2. *it preserves the same zeta carrier and nontrivial zerohood predicate;*
3. *it preserves the same explicit-formula route and same-zero contribution class;*
4. *it preserves the critical normalization by $x^{1/2}$;*
5. *it preserves theorem-bearing target-slot fixation, minimal report evaluability, same-regime fidelity, and act-time finality;*
6. *it preserves $C^0_{\text{RH-EF}}(\rho)$;*
7. *it permits endpoint-resolving non-neutral amplitude governance that is not support-only, not book-keeping, not bridge-complete neutralization, not carrier/domain/interface shift, and not a lawful coequal target role.*

Endpoint determinacy here fixes only the carrier and report role needed for theorem-bearing evaluation. It does not presuppose K13 endpoint exhaustion; K13 is the local theorem that unknown, deferred, unsupported, or hidden-fifth statuses cannot occupy endpoint truth.

Lemma 13.1 (Plural analytic routes are permitted; plural endpoint interiors are not)**.** *K5 does not prohibit plural analytic methods: zero-free regions, density estimates, positivity criteria, explicit-formula variants, smoothing, truncation, pairing, cancellation arguments, finite verification, Hilbert–Polya candidates, random-matrix analogies, or spectral programs. It prohibits only plural endpoint-governing admissible interiors for the same critical-neutral amplitude slot when those interiors assign different endpoint status while preserving the same zero, carrier, route, normalization, and endpoint image.*

Proof. Plural analytic routes may coexist as support, bridge programs, presentations, or scoped completions. If two endpoint interiors agree on all endpoint-bearing acts, the second is bookkeeping. If they disagree while preserving the same zero, carrier, route, normalization, and endpoint image, the same endpoint slot receives two governance verdicts. That disagreement must be resolved by a selector, a domain/interface split, or a second classifier. The first is fresh endpoint authority, the second exits the fixed carrier, and the third is independent amplitude discrimination. □

Lemma 13.2 (Local necessity of K5)**.** *Under $C^0_{\text{RH-EF}}(\rho)$, a weakening that permits two distinct endpoint-governing admissible interiors for the same critical-neutral amplitude slot either collapses to bookkeeping, splits the domain/interface, or creates $D_{\text{amp}}(\rho)$.*

Proof. Let I_1 and I_2 be alleged endpoint-governing interiors for the same critical-neutral amplitude endpoint image. If they agree on all endpoint-bearing acts, their difference is bookkeeping. If they disagree while preserving the same zero, route, normalization, and endpoint image, the disagreement is endpoint-status work on the same carrier. Since it is not neutral occupation, it is independent same-domain amplitude discrimination unless it is reclassified as a domain/interface split. □

Lemma 13.3 (Local necessity of K6)**.** *Under $C^0_{\text{RH-EF}}(\rho)$, endpoint-standing non-neutral amplitude outside admissibility is support-only, carrier/interface shift, or a second endpoint-status gate.*

Proof. If the non-neutral amplitude has no endpoint effect, it is support or bookkeeping. If it has endpoint effect while standing outside the admissibility boundary and preserving the fixed carrier, it governs endpoint status by a second gate. If it avoids that conclusion by changing the zero, carrier, route, normalization, contribution class, or endpoint image, it is carrier/interface shift. These cases exhaust endpoint-standing use outside admissibility. □

Lemma 13.4 (Local necessity of K11)**.** *Under $C^0_{\text{RH-EF}}(\rho)$ plus minimal report evaluability, a theorem-bearing endpoint report that exceeds zerohood, coordinate report, explicit-formula contribution, normalization, bridge status, and amplitude support is unsupported endpoint authority, silent redescription, or no-carrier transfer.*

Proof. Minimal report evaluability attaches a report to the fixed target and classifies it as theorem-bearing, support-level, bookkeeping, or out-of-domain. K11 adds no-overreporting. A coordinate report $\beta \neq 1/2$ cannot be reported as endpoint defeat without the endpoint-use role. A support report $x^{\beta-1/2} \neq x^0$ cannot be reported as contradiction. Cancellation cannot be reported as endpoint-neutral occupation without a bridge. Finite verification cannot be reported as universal neutral occupation. A spectral analogue cannot be reported as the same-zero explicit-formula contribution without carrier preservation. A density estimate cannot be reported as pointwise critical-line readout without the required bridge. If such overreporting remains on the carrier, it is unsupported endpoint authority; if it hides a surrogate as endpoint readout, it is silent redescription; if it imports status from another carrier, it is no-carrier transfer. □

Lemma 13.5 (Local necessity of K13)**.** *K13 does not deny open status, unproved status, finite verification, conditional analytic support, smoothing/truncation artifacts, or incomplete bridge work. It states that those are epistemic, report, or support statuses, not endpoint truth branches on the fixed RH-EF endpoint image.*

Proof. Target-slot fixation says what is being evaluated. K13 says that once the carrier, route, endpoint image, and admissibility boundary are fixed, a theorem-bearing endpoint branch is admitted for endpoint standing or not. Open, deferred, unsupported, finite, conditional, smoothed, or bookkeeping status may describe a proof state or support state. Treating any of them as a third endpoint truth branch destroys determinate same-carrier endpoint use. □

Theorem 13.1 (No strict same-carrier weakening permits endpoint-standing non-neutral amplitude governance)**.** *For a fixed RH-EF instance,*

$$C^0_{\text{RH-EF}}(\rho) \Rightarrow \neg\exists W\, W_{\text{RH-EF}}(W, \rho).$$

Equivalently, no weakening preserving the same zero, carrier, explicit-formula route, contribution class, critical normalization, endpoint image, target-slot fixation, minimal report evaluability, and act-time finality can permit endpoint-standing non-neutral amplitude governance while avoiding support status, bookkeeping, bridge-complete neutralization, carrier/interface shift, lawful coequal target role, and $D_{\text{amp}}(\rho)$*.*

Proof. Assume such a weakening W exists, and let $G^-_{\text{amp}}(\rho)$ be the non-neutral endpoint-governance act it permits. If G^-_{amp} shares the positive admissible interior, it collapses into bridge-complete neutral occupation and no longer preserves endpoint-defeating non-neutral force. If it has a distinct admissible interior, local K5 gives plural endpoint governance, a domain/interface split, or $D_{\text{amp}}(\rho)$. If it has endpoint standing outside admissibility, local K6 gives support-only status, carrier/interface shift, or a second endpoint-status gate. If its report exceeds zerohood, contribution, normalization, bridge, or support, local K11 gives unsupported endpoint authority, silent redescription, or no-carrier transfer. If it is neither admitted nor rejected while still endpoint-resolving, local K13 gives third-status degeneration rather than determinate endpoint use. Finally, if none of these branches applies, the act is same-carrier endpoint-status work that is not neutral occupation, not support, not bookkeeping, not bridge completion, and not carrier/interface shift. By the endpoint-discriminator trichotomy, it is $D_{\text{amp}}(\rho)$, and A+ excludes it. Therefore no such strict same-carrier weakening survives. □

Corollary 13.1 (Strict weakening either preserves endpoint behavior or exits endpoint use)**.** *Every same-carrier weakening of the RH-EF endpoint packet either preserves endpoint behavior extensionally, lowers non-neutral amplitude to raw/support/bookkeeping status, bridge-neutralizes into* $\text{ord}_{1/2}(\rho) = 0$*, changes carrier, route, zero identity, normalization, or endpoint image, or reintroduces* $D_{\text{amp}}(\rho)$*.*

Proof. If the weakening changes no endpoint-bearing verdict, it is endpoint-equivalent. If it changes a verdict while retaining endpoint force, the preceding theorem forces bridge completion, support/bookkeeping collapse, carrier/interface shift, or independent amplitude discrimination. These alternatives exhaust strict same-carrier weakening. □

13.2 Route-locus collapse audit for RH-EF negative use

The route-locus audit prevents endpoint-resolving off-line force from escaping by moving through a familiar analytic route without declaring its endpoint role. Each locus is lawful as mathematics. It is blocked only when it is overreported as same-carrier endpoint defeat or endpoint occupation without the required bridge.

Locus	**RH-EF form**	**Collapse**
L1 Coordinate report	$\beta \neq 1/2$ is lawful coordinate content.	Raw/support unless endpoint-used; endpoint-used coordinate report projects to non-neutral amplitude role.
L2 Explicit-formula support	The contribution x^ρ/ρ, the scale $x^{\beta-1/2}$, and $\mathrm{ord}_{1/2}(\rho) \neq 0$ are lawful EF content.	Support unless overreported; endpoint-standing use enters $O_{\mathrm{nat}}(\rho)$ and D_{amp} routing.
L3 Pairing and cancellation	Functional-equation symmetry, conjugate pairing, smoothing cancellation, and aggregate trace cancellation are lawful bridge material.	Bridge-neutralized if complete; support/bookkeeping otherwise; domain/interface shift if same-zero identity or endpoint image changes; D_{amp} if used as independent endpoint-status work.
L4 Smoothing and truncation	Smoothing kernels, test functions, truncations, and finite-height cutoffs are skin/support unless they alter tensor content.	Skin if presentation-only; support if analytic; carrier/tensor shift if endpoint role changes.
L5 Spectral surrogate	A spectral operator or Hilbert–Polya model may support a bridge.	Support or bridge program; carrier shift if unbridged; positive completion if it yields neutral order on the fixed carrier.
L6 Positivity criterion	A positivity criterion may provide support or an alternative analytic proof route.	Support/bridge unless it yields critical-neutral readout on the fixed carrier.
L7 Numerical verification	Finite-height verification is legitimate evidence and route support.	Support only unless a universal bridge is supplied; no finite-ledger transfer to the universal carrier.

L8 Density estimate	Zero-density information is lawful analytic support.	Support unless it supplies the required universal pointwise bridge; overreporting otherwise.
L9 Partial zero-free region	A zero-free region outside some domain is lawful.	Scoped bridge completion only; support outside scope; no universal transfer.
L10 Carrier, route, or normalization shift	Changing zeta carrier, same-zero identity, prime-trace carrier, normalization, endpoint image, or symmetric outcome carrier is not same RH-EF endpoint use.	Domain/interface shift unless bridged back.
L11 Repair	Later analytic proof, cancellation theorem, zero-free region, spectral construction, or positivity result follows an earlier unbridged endpoint report.	New bridge-complete act; no same-act repair.
L12 Endpoint classifier	An extra classifier decides critical-neutral occupation beside the admitted EF bridge.	Bookkeeping, governance-equivalence, domain/interface shift, or D_{amp}.
L13 Hidden fifth/sixth native role	Endpoint-defeating non-neutral amplitude refuses endpoint-use and discriminator classification.	Exactly $O_{\mathrm{nat}}(\rho)$ under deployment; collapsed in Section 7.

Theorem 13.2 (Route-locus collapse for RH-EF negative use)**.** *Every same-carrier endpoint-resolving off-line RH-EF route enters one of the route loci above. If it enters none, it does no endpoint work and cannot resolve or defeat the endpoint.*

Proof. A negative route must use coordinate displacement, explicit-formula support, pairing or cancellation, smoothing or truncation, a spectral surrogate, a positivity criterion, finite verification, density data, a partial zero-free region, carrier or normalization change, repair, an endpoint classifier, or a hidden native-falsification role. The table covers those ways of changing coordinate, amplitude, carrier, time, locality, selection, or endpoint role. If a route changes none of them and supplies no bridge, it has no endpoint-relevant content. □

14 Denial Burden and Faithful Counterexample Worksheet

The structural-to-EF bridge sharpens the central denial burden. A critic may no longer attack a bare step from off-line coordinate to separator occupation; the manuscript now passes through order-class exactness, B_{ζ}^{EF}, and the bridge-discriminator-domain fork. A successful objection must break one of those links or show that bridge-failing same-carrier non-neutral order can retain endpoint counterforce without either domain shift or independent discriminator work.

14.1 Right-mode denial targets

A standing objection must identify at least one of the following failures:

1. no-autonomous-counterexample-role failure: an official counterexample act defeats the endpoint while retaining official endpoint force and nevertheless not being endpoint use;
2. counterexample-force bridge failure: an official counterexample act defeats the endpoint while not fixing target, carrier, branch, reportability, downstream reuse, branch exclusion, act-time finality, and redescription-stable role;
3. official RH endpoint misidentified;
4. zeta/prime-trace carrier inadequacy;
5. context smuggling of RH or line support;
6. explicit-formula route does not project zerohood to amplitude content;
7. amplitude exactness failure for $x^{\beta-1/2}$;
8. term-isolation failure;
9. autonomous pairing/cancellation-role failure: cancellation affects endpoint status on the fixed RH-EF carrier while being neither support, bridge completion, bookkeeping, carrier/interface shift, nor $D_{\mathrm{amp}}(\rho)$;
10. projection-role content failure;
11. single amplitude interface failure;
12. descriptive/support versus endpoint-standing distinction failure;
13. non-neutral scale standing theorem failure;
14. non-native amplitude theorem failure;
15. endpoint-discriminator trichotomy failure;
16. A+ no-independent closure failure;
17. local reductio annotation or discharge failure;
18. universalization failure;
19. native off-line falsification package failure: a deployed $O_{\mathrm{nat}}(\rho)$ defeats RH on the same fixed zero, route, and normalization while not being endpoint use;
20. expanded native falsification collapse failure: a same-carrier endpoint-defeating off-line role escapes raw/support, bridge-neutralized, bookkeeping, domain/interface shift, and $D_{\mathrm{amp}}(\rho)$;
21. K-node weakening-resistance failure: a weakening W preserves same zero, EF route, normalization, endpoint image, target-slot fixation, minimal report evaluability, and act-time finality while permitting endpoint-standing non-neutral amplitude governance that is not support, bookkeeping, bridge completion, coequal target role, domain shift, or second classifier;
22. route-locus exhaustion failure: an endpoint-resolving negative route avoids coordinate report, EF support, pairing/cancellation, smoothing/truncation, spectral surrogate, positivity, numerical verification, density, partial zero-free region, carrier shift, repair, endpoint classifier, and hidden native role;
23. coequal negative-role exclusion failure: non-neutral amplitude is a lawful coequal occupant of the critical-neutral amplitude image, rather than non-occupation or a shifted symmetric outcome ledger.

14.2 Faithful counterexample worksheet

Target	Successful objection must produce
No autonomous counterexample role	A same-carrier official counterexample act that defeats the endpoint, retains reportable negative endpoint force, licenses downstream reuse, excludes the positive endpoint branch, and nevertheless is not endpoint use.
Counterexample-force bridge	A same-carrier official counterexample act that defeats the endpoint while not fixing target, carrier, branch, standing, reportability, downstream reuse, branch exclusion, act-time finality, and redescription-stable endpoint role.
Context non-smuggling	A clause where $C_{\text{RH-EF}}$ asserts RH, line readout, neutral amplitude occupation, or no-independent closure.
EF adequacy	A nontrivial zeta zero contribution omitted by the model, or a contribution not tied to the zeta carrier.
Amplitude exactness	A zero $\rho = \beta + i\gamma$ whose critical-normalized scale is not governed by $\beta - 1/2$.
Pairing/cancellation	A proof that symmetry or cancellation affects endpoint status on the fixed RH-EF carrier while being neither support, bridge-complete neutralization, bookkeeping, carrier/interface shift, nor $D_{\text{amp}}(\rho)$.
Single interface	A lawful occupant of the critical-neutral amplitude endpoint image that is non-neutral.
Non-explosiveness	A place where the manuscript derives contradiction from OffLineAmpExcl alone.
Non-neutral standing	Endpoint-standing non-neutral amplitude that is support, bookkeeping, carrier shift, repair, surrogate, coequal occupation, and same-carrier endpoint result all at once.
Hinge theorem	Noncritical amplitude status work that is not independent endpoint-status discrimination and not any lawful alternative role.
A+ closure	$D_{\text{amp}}(\rho)$ compatible with $N_{\text{amp}}(\rho)$.
Reductio discharge	A proof that the annotation is an added premise or that contradiction discharges only a strengthened assumption.
Final endpoint	Failure of arbitrary zerohood closure to imply the universal RH statement.
Native off-line falsification package	A deployed $O_{\text{nat}}(\rho)$ that defeats RH on the same fixed zero, EF route, and normalization while not being endpoint use.
Expanded native falsification collapse	A same-carrier endpoint-defeating off-line role outside raw/support, bridge-neutralized, bookkeeping, domain/interface shift, and $D_{\text{amp}}(\rho)$.
K-node weakening resistance	A weakening preserving same zero, EF route, normalization, endpoint image, target-slot fixation, minimal report evaluability, and act-time finality while permitting endpoint-standing non-neutral amplitude governance that is not support, bookkeeping, bridge completion, coequal target role, domain shift, or second classifier.

Route-locus exhaustion	An endpoint-resolving negative route not captured by coordinate report, EF support, pairing/cancellation, smoothing/truncation, spectral surrogate, positivity, numerical verification, density, partial zero-free region, carrier shift, repair, endpoint classifier, or hidden native role.
Coequal negative-role exclusion	A same-carrier RH-EF endpoint image in which non-neutral amplitude is a lawful coequal occupant of the critical-neutral image rather than non-occupation or a shifted symmetric outcome ledger.

15 Hostile-Referee Objections and Replies

Audit 15.1 (This does not prove a zero-free region)**.** *Correct. The proof class is not zero-free-region construction. A zero-free region could be bridge support; it is not required for endpoint-exclusion proof if amplitude separator exclusion succeeds.*

Audit 15.2 (Off-line zeros are analytically meaningful)**.** *Correct. The proof preserves off-line zerohood as raw analytic syntax and preserves the explicit-formula contribution it would have. The excluded role is endpoint-standing non-neutral amplitude separator occupation.*

Audit 15.3 (The real-part coordinate is native)**.** *Correct. The proof does not attack the real-part coordinate. It routes endpoint counterforce through critical-normalized prime-trace amplitude and classifies endpoint-standing non-neutral amplitude as independent discriminator work.*

Audit 15.4 (Amplitude exponents are native analytic data)**.** *Correct as descriptive and support content. They become AASC-relevant only when used as endpoint-standing authority over the fixed critical-neutral amplitude image.*

Audit 15.5 (The neutral amplitude image assumes RH)**.** *No. It defines the positive endpoint image under the official route. It does not assert universal occupation. Occupation is derived only after the local countercase is excluded.*

Audit 15.6 (A+ is doing analytic work)**.** *No. The explicit-formula route supplies amplitude projection. A+ excludes independent endpoint-status discrimination once the role has been identified. It does not supply the explicit-formula scale exponent or a zero-free estimate.*

Audit 15.7 (The proof proves arbitrary propositions)**.** *No. The anti-trivialization theorem blocks arbitrary $\neg P$. The proof requires a fixed endpoint image, official explicit-formula route, projection-role content, and the non-neutral scale standing theorem.*

16 Consolidated Defensive Audit Matrices

The following section records the consolidated audit surface. Every load-bearing transition has a named statement, a dependency class, an anti-misreading guard, and an exact denial burden. The governing rule is: generic unfamiliarity with AASC vocabulary, demands for a different analytic proof class, or surprise at the conclusion do not constitute same-mode objections. A successful objection must identify a failed premise, failed bridge, failed endpoint-role classification, or failed closure theorem on the fixed RH-EF carrier.

16.1 Load-bearing theorem audit table

Audit node	Load-bearing function	Exact denial burden
Official endpoint fixation	Fixes the RH target, branch split, and critical-line readout without asserting RH.	Show that the official endpoint, carrier, or branch split is misidentified.
Context non-smuggling	Makes $C_{\text{RH-EF}}$ a carrier-and-slot object only.	Exhibit a clause where the context asserts RH, line readout, neutral occupation, or no-independent closure.
Official EF route	Declares the explicit-formula route used for endpoint evaluation.	Show that the EF route is not a lawful route from zeta zerohood to prime-trace contribution data.
Endpoint-closure evaluation	Fixes target, carrier, branch slots, and local counterforce discipline.	Show that $\text{OSEval}_{\text{RH-EF}}$ assumes endpoint completeness, either truth branch, or RH.
Kernel forcing	Routes theorem-bearing endpoint use through target adequacy, standing, reference, admissibility, and irreversibility.	Preserve official endpoint use while removing one kernel role without changing domain.
A+ consequence layer	Supplies AMetric boundary, no-selector, no-repair, no-carrier-transfer, and no-independent-discriminator closure.	Show endpoint use on a fixed carrier that is exempt from these consequences.
ATS anchor locking	Fixes zeta, zerohood, prime-trace carrier, and official route as anchors.	Show a lawful endpoint act that changes an anchor while remaining the same endpoint.
ATS tensor locking	Classifies critical-normalized amplitude exponent as tensor content, not notation skin.	Show non-neutral amplitude can be endpoint-erased by presentation alone.
UEAP report ladder	Blocks overreporting from support or coordinate data to endpoint result.	Show support-level amplitude content can be reported as endpoint outcome without endpoint use.
EF model adequacy	Keeps the analytic zero contribution and prime-trace carrier explicit.	Exhibit a nontrivial zeta zero contribution omitted by the EF model.
Term isolation	Allows a single zero contribution to be typed without claiming global dominance.	Show that local contribution typing requires dominance or a global estimate.
Pairing and cancellation	Allows symmetry and cancellation as bridge material, not automatic endpoint-neutrality.	Prove pairing turns a same-zero non-neutral order into endpoint-neutral occupation without proving the positive endpoint.
Positive projection	Identifies critical-line readout with neutral critical-normalized amplitude.	Show $L_{\text{crit}}(\rho)$ does not correspond to neutral order in the EF route.
Negative projection	Identifies off-line readout with non-neutral critical-normalized amplitude.	Show $\neg L_{\text{crit}}(\rho)$ does not yield non-neutral order for the same zero contribution.
Projection-role content	Binds endpoint use of the negative branch to its EF amplitude role.	Show official EF resolution can accept the route while refusing the role content assigned by the route.

Single amplitude interface	Makes the positive endpoint image neutral amplitude occupation only.	Exhibit a lawful occupant of the critical-neutral image that is non-neutral.
Non-explosiveness	Preserves off-line coordinates and support amplitudes outside endpoint force.	Show the proof derives contradiction from OffLineAmpExcl or $x^{\beta-1/2} \neq x^0$ alone.
Descriptive amplitude preservation	Keeps prime-trace scale data lawful as analytic content.	Show the manuscript bans descriptive off-line amplitude data as such.
Endpoint-standing amplitude	Marks the additional use that turns support amplitude into endpoint-status work.	Show endpoint-standing non-neutral amplitude is neither support nor discriminator nor any audited role.
Non-neutral scale standing	Classifies endpoint-standing non-neutral order as same-carrier non-positive status work.	Produce endpoint-standing non-neutral order that does not affect endpoint status.
Non-native amplitude theorem	Separates native analytic data from non-native endpoint authority.	Show native coordinate or amplitude data automatically carries endpoint force without endpoint use.
Endpoint-discriminator trichotomy	Exhausts status work as bookkeeping, governance-equivalent, domain shift, or independent work.	Inhabit a theorem-bearing, endpoint-resolving, same-domain fifth role.
Hinge theorem	Routes noncritical amplitude status work to $D_{\mathrm{amp}}(\rho)$.	Show the live non-neutral endpoint role avoids D_{amp} while retaining counterforce.
No-independent amplitude closure	Gives $N_{\mathrm{amp}}(\rho) \Rightarrow \neg D_{\mathrm{amp}}(\rho)$.	Show an independent amplitude discriminator compatible with the A+ no-independent closure.
Local separator exclusion	Derives contradiction from $\mathrm{AmpSep}^{\mathrm{loc}}(\rho)$, $D_{\mathrm{amp}}(\rho)$, and $N_{\mathrm{amp}}(\rho)$.	Show local separator occupation does not induce D_{amp} or does not trigger N_{amp}.
Anti-trivialization	Blocks arbitrary $\neg P \Rightarrow P$ instantiation.	Instantiate the proof for arbitrary P without an official route, endpoint image, and projection-role theorem.
Exact-complement annotation	Treats the live local assumption as proof-act annotation, not an object premise.	Show the annotation is an added formula in the discharged context.
Local counterforce	Converts only the live exact-complement countercase into local endpoint counterforce.	Show an inert assumption is being promoted, or show a live countercase can block the endpoint without counterforce.
Annotation discharge	Discharges only $[\neg L_{\mathrm{crit}}(\rho)]_i$.	Show the contradiction depends on a strengthened object-level assumption.
Ordinary negative theoremhood	Preserves negative theoremhood as proof legitimacy.	Show the proof rejects negative theoremhood merely because it is negative.
Incompleteness firewall	Makes same-domain incompleteness either endpoint-inert or governed gate work.	Show true-but-unprovable status changes endpoint standing without becoming endpoint-admissibility-relevant.
Route-locus exhaustion	Classifies analytic, surrogate, numerical, repair, selector, and hidden-fifth routes.	Produce a same-carrier endpoint-resolving negative route outside the route table.
Weakening resistance	Blocks strict weaker same-carrier regimes that admit separator governance.	Define a weaker regime preserving carrier and target adequacy while allowing non-neutral endpoint governance without independence.

Final universalization	Moves from arbitrary zerohood to RH.	Show the arbitrary-zero theorem cannot be universally generalized.

16.2 Adversarial same-mode denial catalogue

All denial packets share the same closure rule: if the objection changes endpoint standing, reportability, branch exclusion, downstream reuse, or admissible continuation, then it is endpoint-admissibility-relevant and must survive the kernel, ATS, UEAP, AMetricity, and no-independent-discriminator closure. If it changes none of those features, it is endpoint-inert for the proof.

Denial route	**Required same-mode objection**	**Failure consequence**
Projection failure	EF route does not carry off-line zerohood into non-neutral critical-normalized amplitude.	Breaks branch projection only.
Amplitude adequacy failure	$\beta - 1/2$ is not the endpoint-relevant normalized scale class.	Breaks EF model adequacy.
Term isolation failure	Individual zero contribution cannot be typed without a dominance theorem.	Breaks local contribution typing.
Pairing neutralization failure	Pairing alone makes off-line scale endpoint-neutral without proving RH.	Breaks single-interface role typing.
Support endpoint collapse	Support-level amplitude content can become endpoint-standing without endpoint use.	Breaks UEAP/no-overreporting.
ATS skin authority	Presentation skin carries endpoint force without anchor or tensor change.	Breaks ATS boundary.
UEAP exception	RH-EF permits lower-status report material to be reported as endpoint result.	Breaks report preservation.
Independent-discriminator denial	Separator occupation does endpoint work but does not induce D_{amp}.	Breaks trichotomy or hinge theorem.
A+ consequence denial	Endpoint use does not yield N_{amp}.	Breaks kernel-to-A+ closure.
Raw negation promotion	The proof treats bare $\neg L_{\mathrm{crit}}$ as endpoint use without route and counterforce.	Breaks anti-trivialization audit only if shown.
Annotation premise objection	Reductio annotation is an object-level premise.	Breaks annotation-discharge theorem.
Ordinary theoremhood shield	Official negative resolution retains endpoint force but escapes endpoint-use discipline.	Breaks no-autonomous-counterexample theorem.
Incompleteness rescue	Independence changes endpoint standing without gate work.	Breaks same-domain incompleteness firewall.
Carrier-shift rescue	A surrogate carrier remains the same RH-EF endpoint without bridge.	Breaks fixed-domain identity.
Critical-normalization smuggling	Critical normalization asserts universal neutral occupation.	Breaks context non-smuggling only if a smuggled clause is exhibited.
Final universalization failure	Arbitrary-zerohood closure cannot generalize to RH.	Breaks final endpoint rule.
Hidden fifth role	Endpoint-resolving non-neutral amplitude is neither support, bookkeeping, carrier shift, repair, coequal occupation, nor independent work.	Breaks endpoint-discriminator trichotomy.

Same-domain weakening	A weaker fixed-carrier endpoint regime admits amplitude separator governance.	Breaks weakening resistance.

16.3 Hostile-referee matrix

Hostile objection	**Controlled response in the manuscript**
This is not a zero-free-region proof.	Correct. The proof class is endpoint-exclusion through EF amplitude role projection.
The explicit formula allows off-line terms.	Correct. They are lawful descriptive/support terms until used as endpoint-standing separator work.
The real part is native analytic data.	Correct. The excluded role is endpoint-standing prime-trace scale discrimination, not the coordinate.
The amplitude exponent is native.	Native as analytic content; non-native only as endpoint-standing authority over the critical-neutral image.
Critical normalization assumes RH.	No. It defines the positive endpoint image and does not assert universal occupation.
A single term need not dominate.	The proof uses scale-class role typing, not domination or zero-free estimates.
Pairing can cancel.	Pairing is allowed; endpoint-neutral cancellation requires a bridge. Endpoint-effective cancellation is support, bridge completion, carrier/interface shift, bookkeeping, or $D_{\mathrm{amp}}(\rho)$; it is not an autonomous fifth role.
Finite verification supports RH.	Yes, as finite support, not universal endpoint occupation.
Density estimates support RH.	Yes, as support; scarcity is not absence without a universal bridge.
Hilbert–Polya or positivity could solve RH.	Yes, if bridged to the same carrier; otherwise they are surrogate/support programs.
The proof bans counterexamples.	It bans only endpoint-standing amplitude separator counterforce; raw syntax and support content are preserved until local contradiction.
The proof proves arbitrary propositions.	No. The anti-trivialization theorem requires an official endpoint route, fixed image, and non-neutral scale-standing theorem.
The local reductio adds an assumption.	No. The annotation is a proof-act role attached to the sole discharged assumption.
Ordinary negative theoremhood is forbidden.	No. It is preserved as proof legitimacy; official endpoint use of it is role-classified.
Open status is a third branch.	No. It is epistemic unless it changes endpoint standing; then it is governed gate work.
A+ does analytic work.	No. EF supplies amplitude content; A+ excludes the independent discriminator after the role is identified.
The support matrix is analytic evidence.	No. Corpus support supplies structural AASC locking only.
The EF route is only a reformulation.	It is the declared endpoint route; reformulation becomes role-bearing only under official endpoint use.
The endpoint image is one-sided.	The broad RH outcome carrier is symmetric; the EF endpoint image has neutral occupation and non-neutral non-occupation.
Lean is substituted for proof.	No. Lean is a reproducibility audit of the RH endpoint-role surface.

16.4 Faithful counterexample worksheet and repetition ledger

A critic must produce an object, not a complaint. The cumulative tasks are: find smuggled RH in $C_{\text{RH-EF}}$; find a zero contribution whose normalized exponent is not $\beta - 1/2$; prove pairing makes off-line scale endpoint-neutral without the positive endpoint; inhabit a lawful endpoint-standing non-neutral role outside support, bookkeeping, repair, carrier shift, surrogate substitution, coequal occupation, and independent discriminator work; make D_{amp} compatible with N_{amp}; show the reductio annotation is an object premise; block universal generalization from arbitrary ρ; or define a weaker same-carrier endpoint regime preserving target adequacy while allowing non-neutral endpoint governance.

Load-bearing link	**Primary location**	**Defensive restatement retained**
Context non-smuggling	Official endpoint section	Theorem ladder, anti-circularity audit, worksheet.
Official route fixation	EF endpoint route	Proof spine, projection-role content, denial catalogue.
Kernel forcing	Kernel/A+ replay	A+ theorem, incompleteness firewall, weakening resistance.
ATS locking	Kernel/A+ replay	Corpus ledger, skin authority audit, formalization map.
UEAP report ladder	Kernel/A+ replay	Non-explosiveness, no-overreporting audit, hostile matrix.
Model adequacy	EF model adequacy	Term isolation, pairing closure, route-locus table.
Branch projection	Projection section	Single-interface theorem, endpoint-role content, theorem ladder.
Non-neutral scale standing	Prime-trace neutrality	Hinge theorem, hostile matrix, worksheet.
Discriminator induction	Separator closure	A+ closure, local contradiction, denial table.
Annotation discharge	Local reductio	Anti-circularity audit, proof spine, local theorem.
Universalization	Final corollary	Endpoint corollary, reference integrity, theorem ladder.

17 Conclusion and Final Same-Mode Denial Fork

This manuscript has one nonclassical dependency: official endpoint-defeating counterexample force is theorem-bearing endpoint use. Once that is granted, the remaining proof is target instantiation plus route exhaustion. The conclusion therefore does not hide the decisive commitment. A reader who accepts the endpoint-use status of official counterexample force need only check the RH-EF instantiation, the structural-to-explicit-formula equivalence bridge, the native-counterforce exhaustion, the local reductio discipline, and the A+ no-independent-discriminator closure. A reader who rejects that dependency should attack the no-autonomous-counterexample endpoint-role theorem directly.

The manuscript fixes the official Riemann Hypothesis endpoint, the zeta/prime-trace explicit-formula carrier, the critical-normalized amplitude endpoint image, and the live exact-complement local counter-case discipline. It preserves raw off-line zerohood, descriptive coordinate information, ordinary negative theoremhood, support-level explicit-formula amplitude displacement, symmetry pairing, cancellation analysis, finite numerical evidence, density estimates, positivity programs, and spectral programs in their lawful roles. None of those is excluded merely as content.

The excluded object is narrower: same-carrier endpoint-standing non-neutral amplitude counterforce that is not bridge-neutralized, not support-only, not bookkeeping, not carrier/domain shift, and not lawful coequal occupation of the critical-neutral endpoint image. The structural-to-explicit-formula bridge turns the live off-line countercase into non-neutral critical-normalized order. Native-counterforce exhaustion closes the proposed ordinary-native-falsification fifth role. The no-autonomous-counterexample theorem closes the remaining attempt to keep official counterexample force outside endpoint-use discipline.

Thus the final same-mode denial fork is exact. A critic must break one of the following links: official counterexample force is endpoint use; endpoint use forces the kernel and A+ consequence layer; the structural-to-EF bridge maps line readout to neutral order and off-line readout to non-neutral order; native counterforce has no fifth endpoint role; fixed-domain use excludes carrier shift; or A+ excludes the independent amplitude discriminator. If those links remain standing, the local off-line countercase induces $D_{\mathrm{amp}}(\rho)$ and the same endpoint-use act yields $\neg D_{\mathrm{amp}}(\rho)$. The local assumption discharges, arbitrary zerohood gives critical-line readout, and universal generalization gives RH.

18 Reference Integrity

Classical analytic references fix the zeta object, the Riemann Hypothesis statement, and the explicit-formula background. They do not supply AASC endpoint closure. AASC corpus references fix endpoint-use governance, ATS locking, UEAP report preservation, AMetric boundary discipline, no repair, no carrier transfer, no silent redescription, nonfactorizability, role occupation, and same-domain incompleteness control. They do not supply analytic estimates for ζ. The proof combines the two only at the declared endpoint-route interface.

19 References

1. B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grosse*, Monatsberichte der Berliner Akademie, 1859.
2. H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.
3. E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.
4. A. Ivic, *The Riemann Zeta-Function*, Dover, 2003.
5. E. Bombieri, *Problems of the Millennium: The Riemann Hypothesis*, Clay Mathematics Institute, 2000.
6. A. J. Maley, *A Structural Solution to the Riemann Hypothesis as a Carrier-Preservation Role-Compression Problem*, 2026.
7. A. J. Maley, *P = NP by Exclusion of the SAT Separator Endpoint*, 2026.
8. A. J. Maley, AASC Corpus Control Matrix, project audit matrix, 2026.
9. A. J. Maley, *AASC RH Endpoint Lean Audit*, version v0.1.0, Zenodo, 2026. DOI: `https://doi.org/10.5281/zenodo.20638480`.
10. A. J. Maley, *AASC-RH-Endpoint-Lean-Audit*, GitHub repository, 2026. `https://github.com/somamaley-ux/AASC-RH-Endpoint-Lean-Audit`.

A Lean4 RH Endpoint Audit Appendix

The manuscript proof above remains the primary proof. The Lean4 material in this appendix is audit support for the RH endpoint proof spine used by the AASC prime-trace amplitude route:

$$\mathrm{RH}^{-}_{\mathrm{bare}} \implies \text{official or local counterexample force} \implies \mathrm{AmpSep}^{\mathrm{occ}}(\rho) \implies D_{\mathrm{amp}}(\rho) \implies \bot,$$

followed by pointwise critical-line readout and the official endpoint theorem. The Lean layer is not used as a substitute for the analytic and manuscript proof chain. Its role is reproducibility support for the AASC endpoint-use bridge, the explicit-formula amplitude separator role, the no-independent-discriminator closeout, the status ledger, and the truth-boundary ledger. The archive records this as a standalone RH endpoint audit, released as version `v0.1.0` and archived under DOI `10.5281/zenodo.20638480` [9,10].

A.1 Release, environment, and audit runner

The Lean audit archive referenced by this manuscript is:

```
https://github.com/somamaley-ux/AASC-RH-Endpoint-Lean-Audit
https://doi.org/10.5281/zenodo.20638480
```

The pinned build and audit data recorded by the release are:

Item	Value
Lean toolchain	`leanprover/lean4:v4.28.0`
mathlib requirement	`mathlib`, revision tag `v4.28.0`
Pinned mathlib revision	`8f9d9cff6bd728b17a24e163c9402775d9e6a365`
Lean library target	`MaleyLean` with root `Main`
Main RH audit target	`MaleyLean.Papers.RH.AuditRunners`
Main proof-spine surface	`MaleyLean/Papers/RH/EndpointClosure.lean`
Status ledger	`MaleyLean/Papers/RH/StatusLedger.lean`
Truth-boundary ledger	`MaleyLean/Papers/RH/TruthBoundary.lean`
Focused audit directory	`Checks/Axiom/`
Focused audit runner	`scripts/check-rh-endpoint-audit.ps1`
Recorded endpoint closure	`RHEndpointClosure=100%`
Final internal endpoint	`OfficialRHEndpoint`
Main closeout anchors	`rhCriticalLineReadout_forced`; `rhEndpoint_of_aascContext`

The verification command supplied by the release is:

```
powershell -ExecutionPolicy Bypass -File scripts/check-rh-endpoint-audit.ps1
```

The runner prints the Lean toolchain, prints the pinned mathlib manifest revision, scans the active Lean audit surface for live `axiom`, `sorry`, `admit`, or `unsafe` declarations, builds `MaleyLean.Papers.RH.AuditRunners`, and runs seven focused AASC/RH audit files. The public repository records that the local audit script and GitHub Actions workflow both pass, and that the active RH audit surface has no live project-level `axiom`, `sorry`, `admit`, or `unsafe` declaration [10].

A.2 Declaration locations

The following declaration locations are taken from the release-facing Lean files. They are included to make the manuscript-to-Lean correspondence checkable without treating Lean as a replacement for the manuscript proof. The public source snapshot is compacted in the raw file view, so this appendix records stable module-and-declaration anchors rather than depending on fragile raw-line offsets.

File	Stable declaration groups
`EndpointClosure.lean`	Semantic carrier: `ZetaZeroCarrier`, zerohood, critical-line readout, explicit-formula route, neutral amplitude, and non-neutral amplitude.
`EndpointClosure.lean`	Endpoint surface: `OfficialRHEndpoint`, `RHEndpointUse`, target adequacy, kernel instantiation, and the A^+ consequence layer.
`EndpointClosure.lean`	Branch surface: bridge completeness, off-line branch, bare counterexample content, ordinary counterexample theoremhood, official counterexample force, local counterexample force, and local endpoint use.
`EndpointClosure.lean`	Separator surface: amplitude separator occupation, theorem-level amplitude status discriminator, independent amplitude discriminator, and no-independent amplitude-discriminator closure.
`EndpointClosure.lean`	Status/exhaustion surface: endpoint status, governed endpoint use, negative governed endpoint use, counterexample-use classification, and native-counterforce disposition.
`EndpointClosure.lean`	Main theorem anchors: endpoint-use bridge, no-autonomous-counterexample closure, bridge projection to non-neutral amplitude, discriminator induction, local countercase exclusion, pointwise closeout, and global endpoint closeout.
`StatusLedger.lean`	Audit phase, endpoint-closure summary, main anchors, endpoint-progress ledger, and closeout summaries.
`TruthBoundary.lean`	Truth-boundary ledger separating represented semantic carrier fields from the AASC endpoint-route closeout.
`AuditRunners.lean`	Audit-runner registry, audit file list, endpoint-closure percentage check, and combined runner summary.
`Checks/Axiom/*.lean`	Focused `#print axioms` and `#eval` checks for the AASC foundation and RH endpoint surface.

The principal theorem anchors in `EndpointClosure.lean` are:

```
rhOfficialCounterexampleForce_endpointUse
rhNoAutonomousCounterexampleEndpointRole
rhAutonomousOrdinaryNegativeEndpointRole_collapses
rhLocalCounterexampleForce_localEndpointUse
rhCriticalReadout_iff_neutralAmplitude_of_bridge
rhOffLineBranch_projects_to_nonNeutralAmplitude
rhOfficialCounterexampleForce_projects_to_amplitudeSeparator
rhLocalCounterexampleForce_projects_to_localAmplitudeSeparator
rhAmplitudeSeparator_induces_theoremLevelDiscriminator
rhTheoremLevelDiscriminator_independent
```

```
rhAmplitudeSeparator_induces_independentDiscriminator
rhNoIndependentAmplitudeDiscriminator_excludes_separator
rhEndpointUse_yields_noIndependentAmplitudeDiscriminator
rhNativeCounterforce_hiddenFifthRole_impossible
rhEndpointResolvingNonUse_hiddenFifthCase_impossible
rhGovernedEndpointUse_bivalent
rhNegativeGovernedEndpointUse_has_amplitudeSeparatorStatus
rhCounterexampleForce_impossible_of_noIndependentDiscriminator
rhLocalCountercase_impossible_of_noIndependentDiscriminator
rhCriticalLineReadout_forced
rhPointwiseEndpoint_of_aascContext
rhEndpoint_of_aascContext
rhEndpointClosurePercent_eq
```

The focused audit files named by the runner are:

```
Checks/Axiom/MinimalConditionsForAdmissibleConstructionAxiomCheck.lean
Checks/Axiom/NonDegenerateConstructionAndKernelOfAdmissibilityAxiomCheck.lean
Checks/Axiom/RHEndpointClosureAxiomCheck.lean
Checks/Axiom/RHStatusLedgerAxiomCheck.lean
Checks/Axiom/RHTruthBoundaryAxiomCheck.lean
Checks/Axiom/RHAuditRunnersAxiomCheck.lean
Checks/Axiom/RHFullStackAASCAxiomCheck.lean
```

A.3 Manuscript-to-Lean correspondence

The manuscript-to-Lean symbol correspondence is:

Manuscript symbol or role	Lean-facing declaration
$A_\zeta(\rho)$	`NontrivialZero rho`
$L_{\mathrm{crit}}(\rho)$	`CriticalLineReadout rho`
Official RH endpoint	`OfficialRHEndpoint`
Fixed explicit-formula route	`ExplicitFormulaRoute rho`; packed into `RHExplicitFormulaBridgeComplete rho`
Neutral critical-normalized amplitude	`NeutralPrimeTraceAmplitude rho`
Non-neutral prime-trace amplitude	`NonNeutralPrimeTraceAmplitude rho`
Explicit-formula bridge completeness	`RHExplicitFormulaBridgeComplete rho`
Bare off-line branch	`RHOffLineBranch rho`
Bare counterexample content	`RHBareCounterexampleContent rho`
Ordinary counterexample theoremhood	`RHOrdinaryCounterexampleTheoremhood rho`
Official endpoint-defeating counterexample force	`RHOfficialCounterexampleForce rho`
Local exact-complement countercase force	`RHLocalCounterexampleForce rho`
Endpoint use	`RHEndpointUse rho`

Local endpoint use	`RHLocalEndpointUse rho`
Prime-trace amplitude separator occupation	`RHAmplitudeSeparatorOccupation rho`
Theorem-level amplitude status discriminator	`RHTheoremLevelAmplitudeStatusDiscriminator rho`
Independent amplitude discriminator	`RHIndependentAmplitudeDiscriminator rho`
No-independent amplitude closure	`RHNoIndependentAmplitudeDiscriminator rho`
Positive/separator endpoint bivalence	`RHEndpointStatus`; `rhGovernedEndpointUse_bivalent`
Hidden fifth role closure	`rhNativeCounterforce_hiddenFifthRole_impossible`; `rhEndpointResolvingNonUse_hiddenFifthCase_impossible`
Pointwise critical-line closeout	`rhCriticalLineReadout_forced`
Official endpoint closeout	`rhEndpoint_of_aascContext`
Endpoint closure status	`RHEndpointClosurePercent = 100`

The formal correspondence of the central manuscript chain is:

```
RHOfficialCounterexampleForce rho
  -> RHEndpointUse rho

RHExplicitFormulaBridgeComplete rho
  -> RHOffLineBranch rho
  -> NonNeutralPrimeTraceAmplitude rho

RHExplicitFormulaBridgeComplete rho
  -> RHOfficialCounterexampleForce rho
  -> RHAmplitudeSeparatorOccupation rho

RHAmplitudeSeparatorOccupation rho
  -> RHTheoremLevelAmplitudeStatusDiscriminator rho
  -> RHIndependentAmplitudeDiscriminator rho

RHExplicitFormulaBridgeComplete rho
  -> RHNoIndependentAmplitudeDiscriminator rho
  -> Not (RHOfficialCounterexampleForce rho)

RHExplicitFormulaBridgeComplete rho
  -> RHEndpointUse rho
  -> RHNoIndependentAmplitudeDiscriminator rho
  -> NontrivialZero rho
  -> CriticalLineReadout rho

RHAASCContext -> OfficialRHEndpoint
```

In manuscript terms, this is the checked version of the path from endpoint-defeating counterexample force to endpoint use, from off-line force through the explicit-formula bridge to non-neutral amplitude, from non-neutral endpoint-standing amplitude to the independent amplitude discriminator, and from no-independent closure to contradiction and local discharge.

A.4 Selected source excerpts

The following excerpts are reproduced in manuscript-facing form from `EndpointClosure.lean`. The source uses a semantic carrier rather than a first-principles formalization of the zeta function or the explicit formula.

```
structure ZetaZeroCarrier where
  id : Nat
  nontrivialZerohood : Bool
  criticalLineReadout : Bool
  explicitFormulaRoute : Bool
  neutralAmplitude : Bool

def NontrivialZero (rho : ZetaZeroCarrier) : Prop :=
  rho.nontrivialZerohood = true

def CriticalLineReadout (rho : ZetaZeroCarrier) : Prop :=
  rho.criticalLineReadout = true

def NeutralPrimeTraceAmplitude (rho : ZetaZeroCarrier) : Prop :=
  rho.neutralAmplitude = true

def NonNeutralPrimeTraceAmplitude (rho : ZetaZeroCarrier) : Prop :=
  Not (NeutralPrimeTraceAmplitude rho)

def OfficialRHEndpoint : Prop :=
  forall rho : ZetaZeroCarrier,
    NontrivialZero rho -> CriticalLineReadout rho
```

The endpoint-use and amplitude-separator roles are represented by the following audit-surface declarations:

```
def RHEndpointUse (_rho : ZetaZeroCarrier) : Prop := True

def RHExplicitFormulaBridgeComplete (rho : ZetaZeroCarrier) : Prop :=
  NontrivialZero rho /\ ExplicitFormulaRoute rho /\
  (CriticalLineReadout rho <-> NeutralPrimeTraceAmplitude rho)

def RHOffLineBranch (rho : ZetaZeroCarrier) : Prop :=
  NontrivialZero rho /\ Not (CriticalLineReadout rho)

def RHBareCounterexampleContent (rho : ZetaZeroCarrier) : Prop :=
  RHOffLineBranch rho

def RHOfficialCounterexampleForce (rho : ZetaZeroCarrier) : Prop :=
  RHBareCounterexampleContent rho /\ RHEndpointUse rho

def RHLocalCounterexampleForce (rho : ZetaZeroCarrier) : Prop :=
  RHBareCounterexampleContent rho /\ RHEndpointCarrierFixed rho

def RHAmplitudeSeparatorOccupation (rho : ZetaZeroCarrier) : Prop :=
  RHEndpointUse rho /\ ExplicitFormulaRoute rho /\
  NonNeutralPrimeTraceAmplitude rho
```

The branch-asymmetry and discriminator induction surface is:

```
def RHTheoremLevelAmplitudeStatusDiscriminator
```

```
    (rho : ZetaZeroCarrier) : Prop :=
  RHAmplitudeSeparatorOccupation rho

def RHIndependentAmplitudeDiscriminator
    (rho : ZetaZeroCarrier) : Prop :=
  RHTheoremLevelAmplitudeStatusDiscriminator rho

def RHNoIndependentAmplitudeDiscriminator
    (rho : ZetaZeroCarrier) : Prop :=
  Not (RHIndependentAmplitudeDiscriminator rho)

theorem rhOfficialCounterexampleForce_endpointUse
    {rho : ZetaZeroCarrier} :
  RHOfficialCounterexampleForce rho -> RHEndpointUse rho := by
  intro h
  exact h.2

theorem rhOffLineBranch_projects_to_nonNeutralAmplitude
    {rho : ZetaZeroCarrier} :
  RHExplicitFormulaBridgeComplete rho ->
  RHOffLineBranch rho -> NonNeutralPrimeTraceAmplitude rho := by
  intro hBridge hOff hNeutral
  exact hOff.2
    ((rhCriticalReadout_iff_neutralAmplitude_of_bridge hBridge).2 hNeutral)

theorem rhAmplitudeSeparator_induces_independentDiscriminator
    {rho : ZetaZeroCarrier} :
  RHAmplitudeSeparatorOccupation rho ->
  RHIndependentAmplitudeDiscriminator rho := by
  intro hSep
  exact rhTheoremLevelDiscriminator_independent
    (rhAmplitudeSeparator_induces_theoremLevelDiscriminator hSep)
```

The pointwise and global closeout anchors are:

```
theorem rhLocalCountercase_impossible_of_noIndependentDiscriminator
    {rho : ZetaZeroCarrier} :
  RHExplicitFormulaBridgeComplete rho ->
  RHEndpointUse rho ->
  RHNoIndependentAmplitudeDiscriminator rho ->
  Not (RHLocalCounterexampleForce rho) := by
  intro hBridge hUse hNo hLocal
  exact hNo (rhAmplitudeSeparator_induces_independentDiscriminator
    (rhLocalCounterexampleForce_projects_to_localAmplitudeSeparator
      hBridge hLocal hUse))

theorem rhCriticalLineReadout_forced
    {rho : ZetaZeroCarrier} :
  RHExplicitFormulaBridgeComplete rho ->
  RHEndpointUse rho ->
  RHNoIndependentAmplitudeDiscriminator rho ->
  NontrivialZero rho -> CriticalLineReadout rho := by
  intro hBridge hUse hNo hZero
  by_cases hCritical : CriticalLineReadout rho
  · exact hCritical
  · exact False.elim
      (rhLocalCountercase_impossible_of_noIndependentDiscriminator
        hBridge hUse hNo
        (And.intro (And.intro hZero hCritical) trivial))
```

```
structure RHAASCContext where
  bridgeComplete : forall rho : ZetaZeroCarrier,
    NontrivialZero rho -> RHExplicitFormulaBridgeComplete rho
  endpointUse : forall rho : ZetaZeroCarrier,
    NontrivialZero rho -> RHEndpointUse rho
  noIndependentAmplitudeDiscriminator : forall rho : ZetaZeroCarrier,
    NontrivialZero rho -> RHNoIndependentAmplitudeDiscriminator rho

theorem rhEndpoint_of_aascContext (ctx : RHAASCContext) :
  OfficialRHEndpoint := by
  intro rho hZero
  exact rhPointwiseEndpoint_of_aascContext
    (ctx.bridgeComplete rho hZero)
    (ctx.endpointUse rho hZero)
    (ctx.noIndependentAmplitudeDiscriminator rho hZero)
    hZero
```

A.5 Focused axiom-output table

The focused audit files issue `#print axioms` commands for the foundation anchors and the RH endpoint anchors below. This appendix states the project-level audit claim recorded by the release: the active RH audit surface is scanned for live `axiom`, `sorry`, `admit`, and `unsafe` declarations, and the runner reports no such project-level declarations. This is not the stronger claim that imported Lean, classical logic, quotient, or mathlib foundations are axiom-free.

Focused audit group	Project-level audit status
Endpoint-use bridge anchors	Incomplete-proof and unsafe-token scan closed on the active RH audit surface.
Explicit-formula projection anchors	Incomplete-proof and unsafe-token scan closed on the active RH audit surface.
Amplitude separator and discriminator anchors	Incomplete-proof and unsafe-token scan closed on the active RH audit surface.
No-independent closeout anchors	Incomplete-proof and unsafe-token scan closed on the active RH audit surface.
Hidden-fifth and bivalence anchors	Incomplete-proof and unsafe-token scan closed on the active RH audit surface.
Pointwise and global endpoint closeout anchors	Incomplete-proof and unsafe-token scan closed on the active RH audit surface.
Status-ledger and truth-boundary ledgers	Incomplete-proof and unsafe-token scan closed on the active RH audit surface.
Audit-runner and full-stack AASC/RH checks	Seven focused files are run by the release audit script.

The theorem anchors covered by the focused endpoint-closure audit include:

```
rhOfficialCounterexampleForce_endpointUse
rhNoAutonomousCounterexampleEndpointRole
rhAutonomousOrdinaryNegativeEndpointRole_collapses
rhLocalCounterexampleForce_localEndpointUse
rhCriticalReadout_iff_neutralAmplitude_of_bridge
```

```
rhOffLineBranch_projects_to_nonNeutralAmplitude
rhOfficialCounterexampleForce_projects_to_amplitudeSeparator
rhLocalCounterexampleForce_projects_to_localAmplitudeSeparator
rhAmplitudeSeparator_induces_theoremLevelDiscriminator
rhTheoremLevelDiscriminator_independent
rhAmplitudeSeparator_induces_independentDiscriminator
rhNoIndependentAmplitudeDiscriminator_excludes_separator
rhEndpointUse_yields_noIndependentAmplitudeDiscriminator
rhNativeCounterforce_hiddenFifthRole_impossible
rhEndpointResolvingNonUse_hiddenFifthCase_impossible
rhGovernedEndpointUse_bivalent
rhNegativeGovernedEndpointUse_has_amplitudeSeparatorStatus
rhCounterexampleForce_impossible_of_noIndependentDiscriminator
rhLocalCountercase_impossible_of_noIndependentDiscriminator
rhCriticalLineReadout_forced
rhPointwiseEndpoint_of_aascContext
rhEndpoint_of_aascContext
rhEndpointClosurePercent_eq
```

The `RHEndpointClosureAxiomCheck.lean` file is the focused check for the endpoint closeout. The status-ledger, truth-boundary, audit-runner, and full-stack AASC/RH check files extend that surface from the pointwise route to the recorded release-status ledger and the full reusable AASC foundation stack.

A.6 Audit scope and boundary

The Lean archive formalizes the proof-spine audit surface consumed by this manuscript: the semantic RH carrier, official endpoint use, explicit-formula route standing, critical-line readout, neutral and non-neutral prime-trace amplitude, amplitude separator occupation, independent amplitude-discriminator induction, no-independent closeout, local countercase discharge, endpoint closure percentage, and full-stack audit-runner registry. It also records the truth-boundary ledger separating represented semantic fields from the AASC endpoint closure route.

The Lean archive does not claim a first-principles formalization of analytic number theory. In particular, it does not formalize the analytic theory of $\zeta(s)$, the explicit formula from first principles, zero-free regions, density estimates, positivity criteria, Hilbert–Polya operators, or numerical verification. Those analytic matters are represented by semantic carrier fields. The Lean claim is local and audit-spine oriented: given the represented RH endpoint carrier, explicit-formula bridge completeness, endpoint-use standing, and the no-independent amplitude-discriminator closure, the pointwise critical-line readout and official endpoint object are checked at the proof-spine level.

The expanded native-falsification package, K-node weakening-resistance section, coequal-role exclusion theorem, pairing/cancellation endpoint-role closure, and route-locus collapse audit are manuscript-level defensive expansions of route roles already represented on the RH endpoint surface: counterexample-force use, amplitude separator occupation, discriminator induction, no-independent closure, local countercase discharge, and pointwise critical-line closeout. They do not add analytic number theory formalization requirements and do not alter the Lean proof-spine endpoint.

A challenge to the Lean layer must therefore identify a concrete mismatch between the cited Lean-facing support surface and the manuscript theorem chain. It is not enough to observe that Lean is not itself the analytic proof, because the manuscript and the archive do not claim that it is. The manuscript proof remains the mathematical proof; the Lean archive supplies reproducibility support for the endpoint-use, separator-discriminator, and closeout routing surface.

B Theorem Ladder

1. Bare counterexample content is separated from endpoint use.
2. Ordinary counterexample theoremhood is granted as kernel-internal theoremhood, not rejected for negativity.
3. Official counterexample force is proved to be endpoint use.
4. No autonomous counterexample endpoint role is proved: official endpoint-defeating force cannot stand outside endpoint use.
5. Autonomous ordinary-negative endpoint roles collapse to raw content, support, bookkeeping, ordinary theoremhood without endpoint force, local syntax without endpoint force, or carrier/domain shift.
6. Official counterexample force is shown to satisfy target adequacy and therefore to trigger the kernel and A+ consequence layer.
7. Local exact-complement counterexample force is proved to be local endpoint use without becoming global negative endpoint outcome.
8. RH off-line counterexample force projects to explicit-formula non-neutral order and enters native-counterforce exhaustion.

1. The official RH endpoint is fixed as $\forall\rho(A_\zeta(\rho) \Rightarrow L_{\mathrm{crit}}(\rho))$.
2. The RH-EF context fixes zeta carrier, prime-trace carrier, explicit-formula route, branch slots, and amplitude endpoint image without asserting RH.
3. Official endpoint resolution through the explicit-formula route satisfies target adequacy.
4. Target adequacy forces the kernel; endpoint use on a fixed domain forces A+.
5. ATS locks anchor, tensor, and skin roles; UEAP locks report levels.
6. The context-bound explicit-formula model is adequate for zero-to-amplitude projection.
7. A zero $\rho = \beta + i\gamma$ projects to critical-normalized amplitude exponent $\beta - 1/2$.
8. The structural-to-EF bridge identifies $L_{\mathrm{crit}}(\rho)$ with $\mathrm{ord}_{1/2}(\rho) = 0$ and $\neg L_{\mathrm{crit}}(\rho)$ with $\mathrm{ord}_{1/2}(\rho) \neq 0$.
9. B_ζ^{EF} instantiates the structural carrier-preservation bridge without containing line support by definition.
10. The bridge fork gives: $\mathrm{BridgeComplete}_{\mathrm{EF}}$, $\mathrm{DomainShift}_{\mathrm{EF}}$, or D_{amp}.
11. Critical-line readout projects to neutral amplitude occupation.
12. Off-line readout projects to non-neutral amplitude non-occupation.
13. Projection-role content binds local exact-complement countercase use to non-neutral amplitude content under the official route.
14. Non-neutral amplitude alone is lawful descriptive/support content.
15. Endpoint-standing non-neutral amplitude is same-domain noncritical amplitude status work.
16. Noncritical amplitude status work induces $D_{\mathrm{amp}}(\rho)$.
17. Endpoint use of local amplitude separator induces $N_{\mathrm{amp}}(\rho)$, hence $\neg D_{\mathrm{amp}}(\rho)$.
18. Local amplitude separator occupation is impossible.

19. Exact-complement local countercase induces endpoint counterforce and local separator occupation.
20. The local countercase contradiction discharges $[\neg L_{\text{crit}}(\rho)]_i$.
21. Arbitrary zerohood yields $L_{\text{crit}}(\rho)$.
22. Universal generalization gives RH.

C Anti-Circularity Audit

Potential misreading	Audit response
The paper treats every counterexample as forbidden.	No. Bare counterexample content is not endpoint use. Only official counterexample force or local exact-complement endpoint counterforce enters endpoint-use discipline.
Objects are not acts.	Correct. The object ρ is raw analytic content; offering it as a same-carrier endpoint-defeating counterexample is an act.
There is an autonomous ordinary-negative endpoint role.	No. Corollary 1.1 states that official endpoint-defeating force entails endpoint use; without endpoint use the branch is raw content, support, bookkeeping, ordinary theoremhood without endpoint force, local syntax without endpoint force, or carrier/domain shift.
The paper bans analytic falsification.	No. It classifies endpoint-standing falsification as endpoint use and then applies the same endpoint discipline that governs official resolution.
The proof proves arbitrary propositions.	No. Arbitrary negation lacks official route projection, endpoint counterforce, and endpoint-status work.

Potential circle or misreading	**Audit response**
Context smuggles RH	$C_{\text{RH-EF}}$ fixes carrier and route only. It does not assert line support or neutral occupation.
Structural EF bridge smuggles RH	B^{EF}_{ζ} fixes same-zero, route, report-level, and trajectory conditions. It does not include L_{crit}, $\text{ord}_{1/2} = 0$, or RH as a premise.
Bridge fork bans counterexamples by fiat	The fork allows bridge-complete neutralization and domain shift. Only bridge-failing same-carrier endpoint counterforce yields D_{amp}.
Order class is just relabeled coordinate	$\text{ord}_{1/2}$ is a recoverable asymptotic order invariant of the critical-normalized EF contribution. It is lawful as data and becomes AASC-relevant only under endpoint standing.
Explicit formula smuggles RH	The explicit formula gives zero-to-prime-trace projection; it does not assert $\beta = 1/2$.
Neutral amplitude assumes positivity	Neutral amplitude is the positive endpoint image, not a universal occupation claim.

Off-line coordinates are illegal	Coordinate reports are lawful. Only endpoint-standing separator use is excluded.
Off-line amplitude is illegal	Non-neutral amplitude is lawful descriptive/support content. Endpoint-standing use is the excluded role.
Pairing ignored	Pairing is lawful analytic structure. It does not erase scale-class typing without endpoint-neutral bridge.
Arbitrary negation becomes separator	Blocked by the anti-trivialization theorem.
Reductio annotation is an added premise	The annotation is proof-act role only; it is not an object-level conjunct.
A+ does analytic estimation	A+ excludes independent endpoint-status work after explicit-formula projection identifies the role.
Ordinary negative theoremhood banned	Negative theoremhood is allowed. Official endpoint use of it is role-classified.
Open or independent status is third branch	Open/independent status is epistemic unless it changes endpoint standing; then it is gate work.

D Notation Ledger

Notation	**Meaning**
$A_\zeta(\rho)$	Nontrivial zeta zerohood: $\zeta(\rho) = 0$ and $0 < \mathrm{Re}(\rho) < 1$.
$L_{\mathrm{crit}}(\rho)$	Critical-line readout: $\mathrm{Re}(\rho) = 1/2$.
$\rho = \beta + i\gamma$	Zero decomposed into real and imaginary parts.
$\psi(x)$	Chebyshev prime trace $\sum_{n \le x} \Lambda(n)$.
$E_\psi(x)$	Prime-trace error $\psi(x) - x$.
$\mathcal{E}_\psi(x)$	Critical-normalized trace $(\psi(x) - x)/x^{1/2}$.
$\mathrm{ord}_{1/2}(\rho)$	Critical-normalized amplitude order $\beta - 1/2$.
$D_{\mathrm{ord}}(\rho)$	Nonzero order-class discriminator $\mathrm{ord}_{1/2}(\rho) \neq 0$.
$B_\zeta^{\mathrm{EF}}(\rho)$	Explicit-formula instantiation of the structural bridge B_ζ, without built-in line support.
$\mathrm{BridgeComplete}_{\mathrm{EF}}(\rho)$	Same-zero EF bridge completion yielding endpoint-neutral order.
$\mathrm{DomainShift}_{\mathrm{EF}}(\rho)$	Shift of carrier, route, zero identity, normalization, or endpoint image.
$C_{\text{RH-EF}}$	Fixed RH explicit-formula endpoint context.
$\mathrm{Model}_{\mathrm{EF}}$	Context-bound explicit-formula endpoint model.
OfficialEFRoute	Official evaluation of RH through the explicit-formula zero-to-prime-trace route.
CritAmpOcc	Critical-neutral amplitude occupation.
OffLineAmpExcl	Non-neutral amplitude non-occupation.
EndpointStandingNonNeutralAmp	Endpoint-standing non-neutral amplitude.
$\mathrm{AmpSep}^{\mathrm{loc}}$	Local amplitude separator occupation inside the reductio subproof.
D_{amp}	Independent amplitude endpoint-status discriminator.
N_{amp}	No-independent amplitude discriminator closure.

ATS	Anchor–tensor–skin structural locking discipline.
UEAP	Report-preservation and laundering-exclusion discipline.

E Corpus Support Ledger

This appendix records the selected support rows extracted from the AASC Corpus Control Matrix. The local use of these rows is structural: kernel forcing, UEAP report discipline, ATS decomposition, AMetric no-selector discipline, no repair, no carrier transfer, silent-redescription control, role-occupancy closure, boundary-trace fixation, non-factorizability, and same-domain incompleteness control. None of these rows supplies an analytic zero-location estimate or a new explicit formula.

Local use	Source id	Type	Object	Manuscript use
AMetric boundary	ERK20260527-New_erk_files_May_41605D-THM-023	Theorem	Admissibility constraint on Operator undefinedness	prevents boundary operators from carrying unauthorized endpoint status
AMetric no selector	ERK20260527-New_erk_files_May_41605D-THM-002	Theorem	Admissibility constraint on no-selector/no-parameter boundary authority	no endpoint-bearing selector or metric at boundary
ATS identity lock	ERK20260527-New_erk_files_May_6DC183-LEM-003	Lemma	(Congruence; identity persistence is forced by stable	same-zero/same-route identity under redescription
ATS redescription lock	ERK20260527-New_erk_files_May_6DC183-THM-028	Theorem	i	blocks treating skin changes as tensor content
Bivalence	ERK20260527-New_erk_files_May_ACEB9B-PRO-012	Proposition	Admissibility and bivalence in the typed domain	fail-closed endpoint status on fixed domain
Boundary no-overpromotion	AASC-EX6-THM-10-1	Theorem	Master endpoint non-promotion theorem	endpoint non-promotion
Boundary trace	ERK20260527-New_erk_files_May_045695-EXM-001	Example	Two-token environmental witness	declared trace requirement for continuation loci
Calibration/post-selection	AASC-EX5-COR-11-2	Corollary	Post-selection exclusion	post-selection and after-the-fact exclusion
Endpoint boundary exactness	AASC-EX6-DEF-9-1	Definition	Endpoint-fixed	endpoint/boundary exactness

Local use	Source id	Type	Object	Manuscript use
Governed construction	ERK20260527-New_erk_files_May_35B594-DEF-004	Definition	Governed construction and non-degenerate construction	endpoint use as governed construction
Kernel raw sequence	ERK20260527-New_erk_files_May_35B594-DEF-002	Definition	Kernel-neutral target adequacy	neutral adequacy precondition for non-degenerate endpoint acts
Kernel target adequacy	ERK20260527-New_erk_files_May_35B594-LEM-001	Lemma	Target adequacy forces the kernel roles	kernel forcing for endpoint use
No carriers	ERK20260527-New_erk_files_May_35B594-COR-012	Corollary	No independent carriers beneath standing	blocks independent carrier of endpoint standing
No generators	ERK20260527-New_erk_files_May_EA512E-THM-010	Theorem	no generators	blocks standing creation from failed or lower-status acts
No hidden transport	NOAR-LEM-001	Lemma	No hidden flavor transport	hidden transport analogy for no hidden amplitude route
No repair	ERK20260527-New_erk_files_May_311F0C-THM-009	Theorem	No post-selection repair	blocks later proof repairing earlier unbridged endpoint act
No same-domain totalization	ERK20260527-New_erk_files_May_3D011A-THM-056	Theorem	No internal totalization	same-carrier totalization block
Quotient identity	ERK20260527-New_erk_files_May_4C9E73-THM-021	Theorem	(Quotient-only presentation is insufficient for fixed-scope	redescription orbit vs invariant identity
Rescue route exhaustion	ERK20260527-New_erk_files_May_3D011A-THM-083	Theorem	Rescue-route exhaustion	blocks semantic/reflection/richer-extension rescue
Role occupancy	ERK20260527-New_erk_files_May_DCCC53-THM-003	Theorem	Forced Role Set	licensed realization of role occupation
Same-domain incompleteness	ERK20260527-New_erk_files_May_FBEDF2-THM-007	Theorem	Exhaustion of same-domain incompleteness objection routes	Godel/independence firewall
Silent redescription	ERK20260527-New_erk_files_May_35B594-COR-007	Corollary	No silent redescription	blocks surrogate/normalization replacement of target

Local use	Source id	Type	Object	Manuscript use
Standing conservation	ERK20260527-New_erk_files_May_8B37FD-PRO-005	Proposition	;	standing can be preserved or destroyed, not created
Transport closure	ERK20260527-New_erk_files_May_12778C-PRO-001	Proposition	Explicit mismatch under tensor omission	same-target transport closure
UEAP carrier adequacy	ERK20260527-New_erk_files_May_AF0C6B-CRT-001	Criterion	Carrier Adequacy Test	report-status and carrier adequacy gate
UEAP laundering	ERK20260527-New_erk_files_May_AF0C6B-DEF-004	Definition	Carrier	blocks report promotion from support to endpoint standing

Audit E.1 (Corpus-use boundary)**.** *The support ledger is not an authority import for RH. It is a provenance record for structural constraints replayed locally in the manuscript. A same-mode objection must attack the local use of a support family, not merely observe that the support family comes from the AASC corpus.*

F Consolidated Appendix Defense Capsules

This audit edition keeps the defensive locks in a compact matrix. Each row states the permitted content, the forbidden slide, and the attack point a same-mode objection must use.

Capsule	**Permitted content retained**	**Forbidden slide / attack point**
Context non-smuggling	Carrier, branch slots, route objects.	No RH, line support, neutral occupation, or closure smuggled into $C_{\text{RH-EF}}$.
Official route is not truth	EF route as evaluation interface.	Route fixation does not assert either branch.
$\text{OSEval}_{\text{RH-EF}}$ boundary	Endpoint-closure counterforce discipline.	Not endpoint-complete evaluation or proof availability.
Raw bivalence	Bare local alternatives under $A_\zeta(\rho)$.	No standing from raw truth alone.
Endpoint image vs outcome carrier	Symmetric RH discussion is allowed.	Fixed EF image uses neutral occupation / non-neutral non-occupation.
Kernel forcing	Ordinary endpoint use under any vocabulary.	Kernel is not optional once target adequacy is granted.
A+ consequence	Downstream closure only.	A+ is not a source of analytic zero-location facts.
AMetric boundary	Metrics and ranks inside admitted proof work.	No pre-boundary endpoint selector.
UEAP report ladder	Coordinate and support reports.	No laundering support into endpoint result.
ATS anchor/tensor/skin	Zeta carrier, amplitude exponent, notation.	Skin cannot carry endpoint authority; tensor changes matter.
No carrier transfer	Spectral, positivity, numerical, density, or surrogate programs.	Same endpoint requires a lawful bridge back.
No post-selection repair	Later results may create new acts.	They do not repair an earlier unbridged endpoint act as the same act.
EF model adequacy	Standard explicit-formula zero contribution data.	Model adequacy is not RH and does not assert all zeros are critical.
Amplitude exactness	$x^{\beta-1/2}$ as normalized contribution scale.	Algebraic role content is not a zero-free estimate.
Term isolation	Single contribution typing.	No dominance assertion is used.
Pairing and cancellation	Symmetry and cancellation as bridge material.	Pairing alone does not erase endpoint role.
Smoothing and truncation	Lawful analytic presentation choices.	They are skin unless they alter tensor/carrier content.
Critical-neutral interface	Positive endpoint image.	Defining neutral image does not assert universal occupation.
Off-line amplitude	Descriptive non-neutral scale.	Not a contradiction until endpoint counterforce use.
Projection-role content	Mediates branch-to-amplitude role.	No one-step raw negation to separator occupation.
Support amplitude	Lower-bound or obstruction material.	Lawful unless overreported as endpoint outcome.
Endpoint-standing non-neutral amplitude	Live counterforce role.	It is status work, not mere support.

Native data as authority	Native coordinates and scales remain valid.	Native data are non-native only as endpoint-standing authority.
No hidden RH-equivalent estimate	No zero-free, density, or positivity theorem assumed.	The proof class is not disguised analytic estimation.
Amplitude separator	Endpoint-standing non-neutral occupation.	Not support, bookkeeping, repair, carrier shift, or coequal image occupation.
Trichotomy	Bookkeeping, governance-equivalent, domain shift, or independent work.	No hidden fifth endpoint role.
D_{amp} induction	Noncritical endpoint-status work.	Descriptive labels alone do not trigger D_{amp}.
N_{amp} closure	Endpoint use of separator role.	A+ is applied only after endpoint use.
Local contradiction	$D_{\text{amp}} \wedge \neg D_{\text{amp}}$.	Contradiction is local to endpoint-standing separator occupation.
Annotation discipline	Exact-complement proof-act role.	Annotation is not a premise.
Temporary local standing	Live reductio countercase only.	No global negative endpoint outcome is inferred.
Endpoint counterforce	Branch used to block endpoint closure.	Narrower than truth, syntax, or support.
Discharge	Sole assumption $[\neg L_{\text{crit}}(\rho)]_i$.	Discharge does not target a strengthened conjunction.
Universalization	Arbitrary nontrivial zerohood.	Not finite verification or local-only theorem.
Ordinary negative theoremhood	Negative proofs as proof legitimacy.	Not banned by polarity.
Official negative resolution	Official endpoint use when used to resolve endpoint.	No theoremhood shield against endpoint-role discipline.
Incompleteness firewall	External proof-system facts.	Same-domain standing effects become governed gate work; otherwise inert.
Open status	Epistemic status.	Not a third endpoint branch.
Route-locus exhaustion	Analytic and surrogate routes.	Endpoint-resolving route must enter the audit table.
Weakening resistance	Alternative formal vocabularies.	Same-carrier weakening cannot admit separator governance without independent work.
Faithful counterexample burden	Concrete object required.	Complaints, wrong-mode demands, and vocabulary refusal are insufficient.

G Transition, Formalization, and Route Audit

The following matrix keeps the transition-by-transition and route-certificate checks while removing repeated prose. The Lean-facing appendix remains Appendix A; the present table is the manuscript-side route map.

Code	Transition	Authorized by	Misreading blocked

T1	$A_\zeta(\rho) \Rightarrow L_{\mathrm{crit}}(\rho) \vee \neg L_{\mathrm{crit}}(\rho)$	Raw bivalence under fixed zerohood.	Standing from truth.
T2	$\neg L_{\mathrm{crit}}(\rho) \Rightarrow \beta \neq 1/2$	Critical-line readout.	Coordinate illicitness.
T3	$\beta \neq 1/2 \Rightarrow x^{\beta-1/2} \neq x^0$	EF amplitude exactness.	Contradiction from scale alone.
T4	Off-line amplitude to projection-role content.	OfficialEFRoute, $\mathrm{Model}_{\mathrm{EF}}$, B_ζ^{EF}.	One-step separator typing.
T5	Projection-role content plus counterforce to endpoint-standing non-neutral amplitude.	Endpoint-standing criterion.	Support promoted to endpoint.
T6	Endpoint-standing non-neutral amplitude to noncritical status work.	Non-neutral scale standing theorem.	Native data banned.
T7	Noncritical status work to D_{amp}.	Endpoint-discriminator trichotomy and hinge theorem.	Hidden fifth role.
T8	$\mathrm{AmpSep}^{\mathrm{loc}} \Rightarrow$ EndpointUse.	Local separator occupation definition.	Support called endpoint use.
T9	Endpoint use to N_{amp}.	Kernel and A+ closure.	A+ applied before use.
T10	$N_{\mathrm{amp}} \Rightarrow \neg D_{\mathrm{amp}}$.	No-independent amplitude closure.	Labels banned as labels.
T11	$D_{\mathrm{amp}} \wedge \neg D_{\mathrm{amp}} \Rightarrow \perp$.	Classical contradiction.	None.
T12	Contradiction discharges $[\neg L_{\mathrm{crit}}]_i$.	Annotation-discharge rule.	Strengthened discharge.
T13	Arbitrary zerohood to universal RH.	Universal generalization.	Finite-only proof.

Route locus	**RH-EF form**	**Disposition**
Coordinate report	$\mathrm{Re}(\rho) \neq 1/2$.	Lawful analytic data; not endpoint standing.
EF support	$x^{\beta-1/2} \neq x^0$.	Lawful support; not contradiction.
Endpoint separator	Non-neutral amplitude used as counterforce.	Routes to D_{amp} and is excluded by N_{amp}.
Symmetry orbit	$\rho, 1-\rho, \bar{\rho}, 1-\bar{\rho}$.	Carrier structure; no same-zero neutral occupation without bridge.
Pairing/cancellation	Paired terms cancel or oscillate.	Bridge if proved; otherwise support.
Spectral surrogate	Hilbert–Polya-style object.	Support or bridge program; not same endpoint without carrier identification.
Positivity criterion	Equivalent or proposed criterion.	Bridge if fully proved; surrogate if not.
Numerical verification	Finite-height checking.	Finite support only.
Density estimate	Scarcity or proportion result.	Support; absence must be proved.
Zero-free region	Partial region exclusion.	Support unless it covers all off-line possibilities.
Carrier shift	Replacement domain.	Not the same RH-EF endpoint.
Repair	Later proof validates earlier report.	New act; no same-act repair.

Hidden fifth role	Endpoint-resolving but non-governing off-line amplitude.	Impossible by trichotomy.

A right-mode objection must attack context adequacy, context non-smuggling, official route fixation, EF model adequacy, amplitude exactness, term isolation, pairing closure, projection-role content, single amplitude interface, non-explosiveness, endpoint-standing criterion, non-neutral scale standing, non-native amplitude theorem, endpoint-discriminator trichotomy, A+ no-independent closure, exact-complement annotation discipline, annotation discharge, or universalization. Demanding a zero-free region, density estimate, or spectral operator is wrong-mode unless the objection also explains why endpoint-exclusion is not a legitimate proof class.

H Detailed Load-Bearing Proof Capsule Index

Capsule	**Role retained**	**Denial burden**
Target fixation without conclusion	Fix official RH-EF target and branches.	Show RH is asserted by target fixation.
Official EF route versus analytic proof	Separate endpoint route from zero-free estimate.	Show the route is analytically invalid.
Same-zero identity	Preserve identity of ρ through projection.	Show carrier substitution or zero replacement.
Smoothing/truncation skin	Keep presentation choices subordinate.	Show skin changes endpoint tensor.
Critical normalization	Define neutral interface.	Show normalization smuggles RH.
Amplitude exactness	Compute normalized scale role.	Show exponent mismatch.
Positive projection	Critical line gives neutral occupation.	Break critical-to-neutral equivalence.
Negative projection	Off-line gives non-neutral order.	Break off-line-to-non-neutral equivalence.
Projection-role content	Bind official route to endpoint role.	Refuse route content while retaining route force.
Descriptive coordinate preservation	Keep coordinates lawful.	Show coordinates are banned.
Descriptive amplitude preservation	Keep amplitude content lawful.	Show support content is banned.
Endpoint-standing criterion	Add endpoint force as separate role.	Show support alone is endpoint standing.
ATS anchor lock	Fix zeta/prime-trace carrier.	Show anchor drift remains same endpoint.
ATS tensor lock	Fix amplitude exponent as content.	Show tensor can be skin.
ATS skin lock	Remove notation-only authority.	Show notation carries endpoint force.
UEAP no-overreporting	Block support-to-endpoint laundering.	Show lower status can report endpoint result.
No carrier transfer	Forbid unbridged surrogate authority.	Show surrogate is same carrier without bridge.
No repair	Fix act-time finality.	Show later repair preserves same endpoint act.

Pairing support	Allow symmetry/pairing.	Show it grants endpoint-neutral occupation.
Cancellation support	Allow analytic cancellation.	Show cancellation erases endpoint role automatically.
Single amplitude interface	Positive image is neutral amplitude.	Exhibit non-neutral occupant of that image.
Coequal carrier separation	Separate symmetric outcome carrier.	Show non-neutral branch is coequal image occupation.
Non-native amplitude theorem	Classify endpoint authority.	Show native data retain endpoint force outside endpoint use.
Discriminator induction	Noncritical status work induces D_{amp}.	Inhabit non-independent endpoint work.
No-independent closure	Endpoint use yields N_{amp}.	Make D_{amp} compatible with closure.
Local reductio role	Live exact-complement assumption only.	Show inert assumption is promoted.
Annotation discharge	Discharge original assumption.	Show strengthened-assumption discharge.
Anti-trivialization	Require RH-EF route and image.	Instantiate arbitrary P.
Ordinary theoremhood	Preserve negative theoremhood.	Show polarity is treated as defect.
Incompleteness firewall	Inert or governed.	Show standing-changing incompleteness avoids gate work.
Open status	Epistemic only.	Make open status a third branch.
Route-locus exhaustion	Close analytic/surrogate routes.	Produce route outside table.
Weakening resistance	No strict same-carrier weakening.	Build weaker regime admitting separator governance.
Corpus support boundary	Structural support only.	Treat corpus support as analytic zero evidence.
Reference integrity	Separate analytic and AASC citations.	Show citation class supplies hidden theorem.

I Hostile-Referee Simulation Matrix

Objection	**Reply retained**
This still does not prove a zero-free region.	Correct; the proof class is endpoint-exclusion, not zero-free-region construction.
Off-line zeros are ordinary analytic objects.	Granted as raw or support content; excluded only as endpoint-standing separator occupation.
The amplitude exponent is native analytic data.	Granted descriptively; endpoint-standing authority is role-classified.
The proof assumes the critical scale.	It defines the positive endpoint image and later derives occupation.
Pairing could cancel the term.	Pairing is bridge material, not automatic endpoint-neutral occupation.
The EF route is an equivalent reformulation.	Official route use fixes endpoint-role content; support-only use is harmless.
A+ is doing analytic work.	EF supplies analytic projection; A+ supplies fixed-domain discriminator closure.

The argument proves any proposition.	Anti-trivialization requires the RH-EF route, image, projection, and non-neutral standing theorem.
Ordinary negative theoremhood is forbidden.	No; only official endpoint use or local endpoint counterforce is role-classified.
Independence could create a third status.	Third status is epistemic or gate work; gate work is governed.
Numerical verification supports RH.	Yes, but finite support is not universal endpoint occupation.
A spectral operator may solve RH.	Yes if bridged to the same carrier; otherwise a surrogate program.
A positivity criterion could solve RH.	Yes if fully proved and bridged; otherwise support.
Density estimates almost prove RH.	They are support; endpoint closure requires absence or a valid bridge.
The proof is too formal.	Formality is not an objection unless a theorem, dependency, or bridge fails.
The context object contains too much.	Context non-smuggling lists what it does not assert.
Projection-role content is artificial.	It is the accepted route's endpoint role, not an extra premise.
The local annotation adds a premise.	The annotation is metaproof role notation; the sole discharged formula is unchanged.
Neutral amplitude is arbitrary.	It is the critical-normalized EF readout of the positive branch.
The corpus matrix is not an analytic proof.	Correct; it is structural support only.

J Detailed Route-Certificate Audit

Certificate	**Question certified**	**Failure mode if absent**
Status coordinate certificate	Which status is being assigned?	Coordinate/support status slides into endpoint status.
Readout-slot certificate	Which endpoint readout is fixed?	Critical-line readout is replaced or blurred.
Occupation certificate	What occupies the endpoint image?	Non-neutral amplitude is treated as coequal occupation.
Carrier-transfer certificate	Is the same zeta/prime-trace carrier preserved?	Surrogate, spectral, positivity, or numerical carrier shift.
Temporal certificate	Is this the same act at act-time?	Post-selection repair.
Locality certificate	Is the role global or local reductio use?	Local countercase becomes official negative endpoint outcome.
Selection certificate	Does a rank, metric, or preferred encoding decide endpoint status?	AMetric no-selector violation.
Pairing certificate	Does pairing prove endpoint-neutrality or merely support it?	Cancellation overreported as endpoint result.
Projection certificate	Is the branch content bound by the EF route?	Raw negation becomes separator without mediation.
Endpoint-use certificate	Is the act theorem-bearing endpoint use or support only?	A+ applied too early or endpoint force denied too late.